



Advances in intelligent and autonomous greenhouse systems: A comprehensive review of internet of things, artificial intelligence, and robotics integration

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ARTICLE INFO

Keywords:

Smart greenhouse
Advanced technologies
Internet of things
Machine/deep learning
Wireless sensor network
Robot

ABSTRACT

Arid and hyper-arid regions are experiencing continual losses of arable land due to environmental degradation, climate change, rapid population growth, and industrial expansion. Greenhouses equipped with advanced monitoring and control technologies offer a promising strategy to sustain and enhance agricultural productivity in these fragile environments. Recent studies reveal a sharp increase in the development of greenhouse monitoring systems; however, their practical application remains limited due to integration complexities, data scarcity, and computational constraints. This review synthesizes over 100 studies published since 2021, focusing on the application of technologies such as the Internet of Things (IoT), machine learning (ML), deep learning (DL), and robotics for yield prediction, disease detection, and plant growth monitoring. All reported datasets, artificial intelligence (AI) models, and their corresponding performance metrics were critically examined. Overall, this review identifies, categorizes, and evaluates current and emerging trends in intelligent greenhouse monitoring and control systems to advance research and innovation in this field. Technologies such as IoT, ML/DL, and robotics are recognized as key innovations that advance unmanned greenhouse management while improving energy and water efficiency. In addition to analyzing cutting-edge technologies in the greenhouse sector, this review also addresses persistent challenges—including multimodal sensor fusion, model adaptability, limited data availability, and computational constraints—along with future trends and research opportunities. This study highlights the significant potential of IoT, ML/DL, and robotics to accelerate the global development of data-driven, intelligent greenhouse applications.

1. Introduction

1.1. Importance of greenhouses

Arable land is steadily declining because of industrialization, rapid population growth, climate change, and environmental pollution [1]. Modern agriculture faces mounting challenges arising from increasing global food demand, labor shortages, climate variability, and the urgent need for sustainable resource management [2]. The global population is projected to reach 10 billion by 2050 [3] and to exceed 11.2 billion by 2100 [4]. As global food requirements rise and both arable land and agricultural labor become increasingly scarce, traditional farming methods are becoming less viable. To meet future global food demands,

production must approximately double by 2050 [5]. Therefore, a paradigm shift toward resilient and sustainable agricultural systems is imperative in the agriculture industry [6,7]. Adopting innovative agricultural technologies is therefore essential to enhance productivity and ensure food security [8]. Controlled Environment Agriculture (CEA) refers to advanced agricultural practices that create optimal growing conditions to protect plants and enhance productivity [9]. In modern agriculture, CEA—which includes techniques like indoor vertical farming and greenhouse cultivation—has become extremely important [10]. These systems aim to maximize resource efficiency through precise regulation of temperature, humidity, and nutrient delivery, resulting in lower water and fertilizer consumption compared to conventional open-field farming [11]. Research has shown that controlled

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<https://doi.org/10.1016/j.atech.2025.101670>

Received 27 August 2025; Received in revised form 11 November 2025; Accepted 23 November 2025

Available online 25 November 2025

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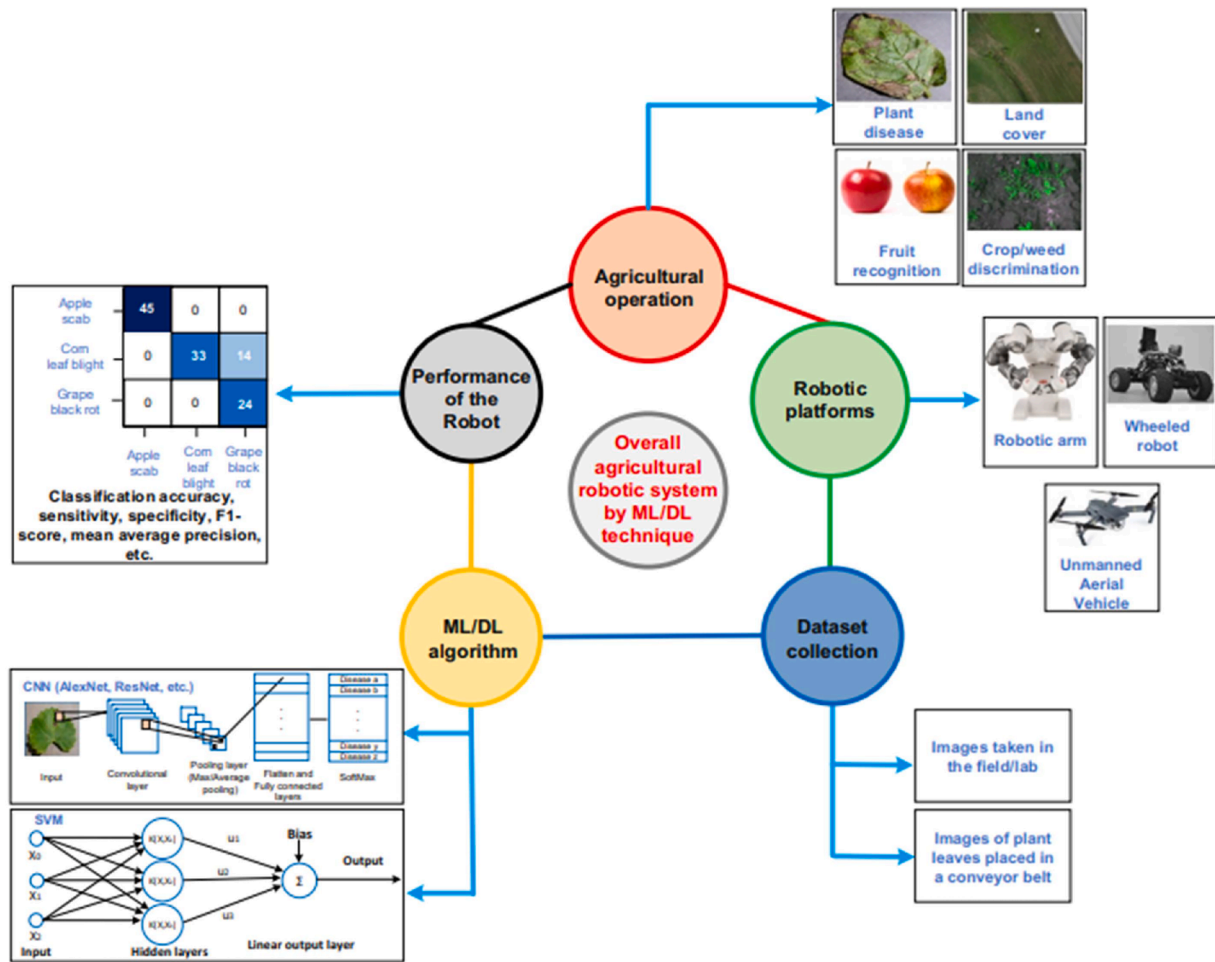


Fig. 1. Advanced technologies in agriculture [33].

environments can greatly improve crop quality, leading to higher marketable yields and greater nutritional value [12]. Among various CEA systems, greenhouses have received considerable attention for their rapid development and contribution to sustainable production [13]. Greenhouse cultivation offers a viable solution by providing regulated conditions that optimize plant growth, yield, and resource use efficiency [10]. A greenhouse is an agricultural structure designed to maintain controlled microclimate conditions, thereby extending the growing season for diverse crops [14].

1.2. Greenhouse technology

Greenhouse technology is among the most recent innovations enhancing agricultural productivity. It enables year-round crop cultivation and is regarded as one of the most promising approaches for sustainable food production [15]. Traditionally, greenhouse monitoring and management were conducted manually under the supervision of farmers, requiring a substantial labor input [16]. By precisely regulating internal environmental parameters such as temperature, humidity, and light according to crop requirements, greenhouses improve both the quality and efficiency of production. However, many conventional greenhouses in arid and developing regions still face challenges such as inadequate environmental control and poor regulation of key parameters (Romano et al., 2020). In recent years, the growing adoption of smart farming practices has accelerated the development of greenhouse farming. Smart greenhouse systems integrate information and communication technologies (ICT) to automate environmental monitoring and control, thereby reducing labor dependence while improving crop yield

and quality. The implementation of intelligent greenhouse technologies has shown clear potential for increasing both the quantity and quality of agricultural output [9].

The promotion of technologically advanced greenhouses, development of intelligent decision-support systems, and adoption of intensive yet sustainable production methods—capable of protecting crops from adverse weather and pest infestations—have become essential trends in modern agriculture [17]. Recent literature provides comprehensive analyses of greenhouse systems, addressing aspects such as design parameters, thermal modeling and simulation, microclimate-controlling technologies, and covering materials [18]. Additional studies have focused on evaluating greenhouse structural design elements, including various shapes [19]. Cuce et al. [20] examined advanced approaches to greenhouse climate control and energy-efficient technologies. Comparative assessments indicate that properly upgrading conventional greenhouses can achieve up to 80 % energy savings, whereas approximately 50 % of production costs in traditional systems are attributed to energy consumption [21]. However, transitioning from traditional to smart greenhouses requires the integration of sophisticated control algorithms, which are considered a crucial component of greenhouse automation [22]. Current research also explores advanced control theories [23] and environmental monitoring through wireless sensor networks (WSNs) and IoT [24].

Through the adoption of cutting-edge technologies that maximize resource efficiency, minimize environmental impact, and enhance food production, the agricultural sector plays a critical role in addressing these global issues. Among these innovations, greenhouse cultivation systems have gained significant attention for their ability to establish

controlled environments that promote higher yields, enhanced crop quality, and reduced water consumption [25]. One effective strategy to reduce the effects of a changing agricultural landscape is CEA, which enhances biosecurity, mitigates pest and drought risks, enables year-round crop production, and supports localized food production in urban areas [26]. As efficient food production is vital to achieving sustainable energy development, energy-saving greenhouse technologies have become increasingly important [27]. Consequently, greenhouses have garnered considerable attention within the agricultural sector for their remarkable technological advancements and developmental progress [28].

1.3. Greenhouse advancements

Recent developments and modifications in greenhouse technology have led to the adoption of science-based solutions to optimize plant production. Modern automatic greenhouse systems employ a variety of sensors and actuators to enable real-time monitoring and control with minimal human input [29]. The integration of intelligent digital monitoring and control systems has elevated greenhouses to meet the high commercial standards demanded by modern agriculture. Understanding and managing greenhouse environmental parameters allow more informed and precise decisions for optimal plant growth [29]. Furthermore, incorporating AI algorithms enhances both the accuracy and cost-efficiency of greenhouse environmental control systems. García-Lesta et al. [30] introduced a novel WSN that uses perpetual motes, which are low-power, energy-harvesting sensor nodes operating autonomously for extended periods. Although initially designed to monitor terrestrial snail activity, this innovation represents a major improvement for greenhouse systems. By integrating such self-sustaining sensor nodes, key greenhouse parameters—such as temperature, humidity, soil moisture, and light intensity—can be monitored continuously and without maintenance. These energy-efficient technologies guarantee optimal growing conditions by facilitating real-time data collection and remote control. According to García-Lesta et al. [30], this development contributes to the creation of intelligent, self-sufficient, and sustainable greenhouses that operate with minimal human intervention.

However, greenhouses are undergoing a transformative shift driven by advances in information technology and innovative decision support systems [17]. Modern greenhouses have evolved from simple covered structures into intricate technological advanced facilities designed to maximize agricultural productivity at minimal cost [31]. In the near future, they are expected to become more self-sufficient and adaptable, enhancing both sustainability and scalability. Greenhouse management and control techniques therefore remain central to contemporary agricultural research for achieving high yields while minimizing energy consumption and production costs [29].

One innovative approach to addressing these issues is the integration of robotics into greenhouse systems. Robotics can enhance precision agriculture by enabling data-driven decision-making, automating labor-intensive operations, and supporting real-time environmental and plant health monitoring [32]. Consequently, many repetitive and precision-dependent operations are being automated through the development of advanced robotic systems. By combining robotics with sensor technologies and AI, greenhouses can manage plant health, climate, and crop yields more accurately, consistently, and efficiently (Bechar & Vigneault, 2016). Robotic platforms equipped with vision systems, environmental sensors, and manipulators facilitate autonomous navigation, real-time data collection, and targeted activities like spraying and harvesting (Kalaitzaki et al., 2021).

Fig. 1 illustrates a typical workflow for implementing a DL or ML algorithm on a field robot. The process begins by identifying the specific agricultural task (e.g., disease scouting, fruit picking, or weed detection). A suitable robotic platform is then selected to navigate the target environment and carry the necessary sensors, which may include red-

green-blue/depth-near-infrared (RGB/D-NIR) cameras, light detection and ranging (LiDAR) units, or multispectral imagers. During operation, the robot collects raw sensor data to generate a labeled dataset that captures real-world diversity in plant phenotypes, lighting, and occlusions. This dataset is then used to (i) develop a suitable ML/DL model (e.g., convolutional neural network (CNN), transformer, or object detector) and (ii) train the model offline by adjusting hyperparameters and loss functions. Once trained, the model is either integrated into the robot's onboard computer or deployed via edge or cloud computing to enable real-time predictions during field operations. Finally, performance metrics such as classification accuracy, F1-score, detection rate, false-positive/negative rates, and processing latency are used to evaluate system performance. These developments contribute to the realization of reliable and autonomous agricultural robots through iterative hardware and computational refinements.

Despite rapid advancements in greenhouse technologies particularly in automation through sensors, AI, and robotics—the current body of research lacks a comprehensive and integrative review that evaluates how these technologies interact, complement each other, and collectively contribute to optimizing greenhouse productivity, sustainability, and cost-efficiency. Most existing studies examine these technologies in isolation or emphasize technical descriptions without addressing their combined practical implementation, interoperability challenges, and real-world deployment constraints in smart greenhouse systems.

Overall, the primary goal of this study was to conduct a comprehensive investigation into the significance, scope, and potential of Agricultural 4.0 research in the context of greenhouse systems. Accordingly, this study aimed to: 1) systematically review and categorize current technological advancements in greenhouse automation, including sensor systems, AI algorithms, and robotic platforms; 2) evaluate the levels of integration among AI, robotics, and environmental sensing technologies in modern greenhouse systems; 3) identify the limitations, interoperability issues, and challenges associated with deploying integrated intelligent systems in real-world greenhouse environments; 4) propose a conceptual framework or roadmap for the next generation of smart, autonomous, and sustainable greenhouse systems, based on the synthesis of current research trends; and 5) highlight future research directions and opportunities, especially in the context of Agricultural 4.0, to advance fully autonomous and efficient greenhouse operations.

2. Methodology of work (Review scope)

A comprehensive literature search was conducted across major scientific databases—including Google Scholar, Springer, Elsevier, Science Direct, Taylor & Francis, Nature, and IEEE—to identify over 100 relevant research articles published between 2020 and 2025. Articles were filtered using specific keywords related to greenhouses ("greenhouse", "controlled environment", and "indoor farm"), technology ("greenhouse technology", "Internet of Things", and "smart farming", and "IoT"), and AI ("AI", "machine learning", "ML", "deep learning", "DL", "robot", "robotic system", "agricultural robot", and "autonomous robot"). Although many studies focused on broader agricultural contexts, a clear gap was observed in research exclusively dedicated to greenhouse farming. Recognizing that greenhouse environments remain underexplored, this study contributes by systematically reviewing and analyzing advanced technology applications within this specific domain. The review highlights different state-of-the-art methods and elucidates their applicability to greenhouse agriculture.

Several studies have investigated the use of robotics, ML/DL, and IoT in agriculture. However, there remains a clear gap in literature regarding the comprehensive application of these technologies, especially in greenhouse farming. This study seeks to address this gap by providing an in-depth review of advanced technological approaches—particularly IoT, ML/DL, and robotics—and their integration in greenhouse environments. The main contributions of this review are as

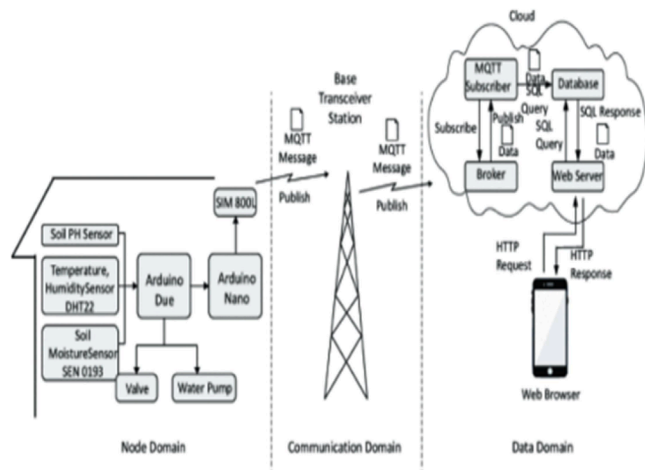


Fig. 2. Architecture of the smart greenhouse for environmental engineering applications.

follows:

(1) It addresses a major research gap by presenting a focused and comprehensive review of IoT-enabled, robotic, and ML/DL-based technologies applied in greenhouse systems. (2) It examines and synthesizes current methods, potentials, limitations, and future research needs for advanced technologies in key greenhouse application domains, including yield estimation, disease/health detection, and growth monitoring. (3) It highlights real-world implementation issues, summarizes emerging trends, and provides visual illustrations to support understanding. Overall, this review consolidates recent developments, highlights critical research gaps, and outlines future directions for intelligent and sustainable agriculture.

3. IoT-Driven monitoring and control of the greenhouse microclimate

IoT has transformed environmental management by integrating everyday objects into intelligent systems such as smart greenhouses. Through smart sensing, data transfer, and long-range wireless communication, it enables farmers to monitor and regulate greenhouse

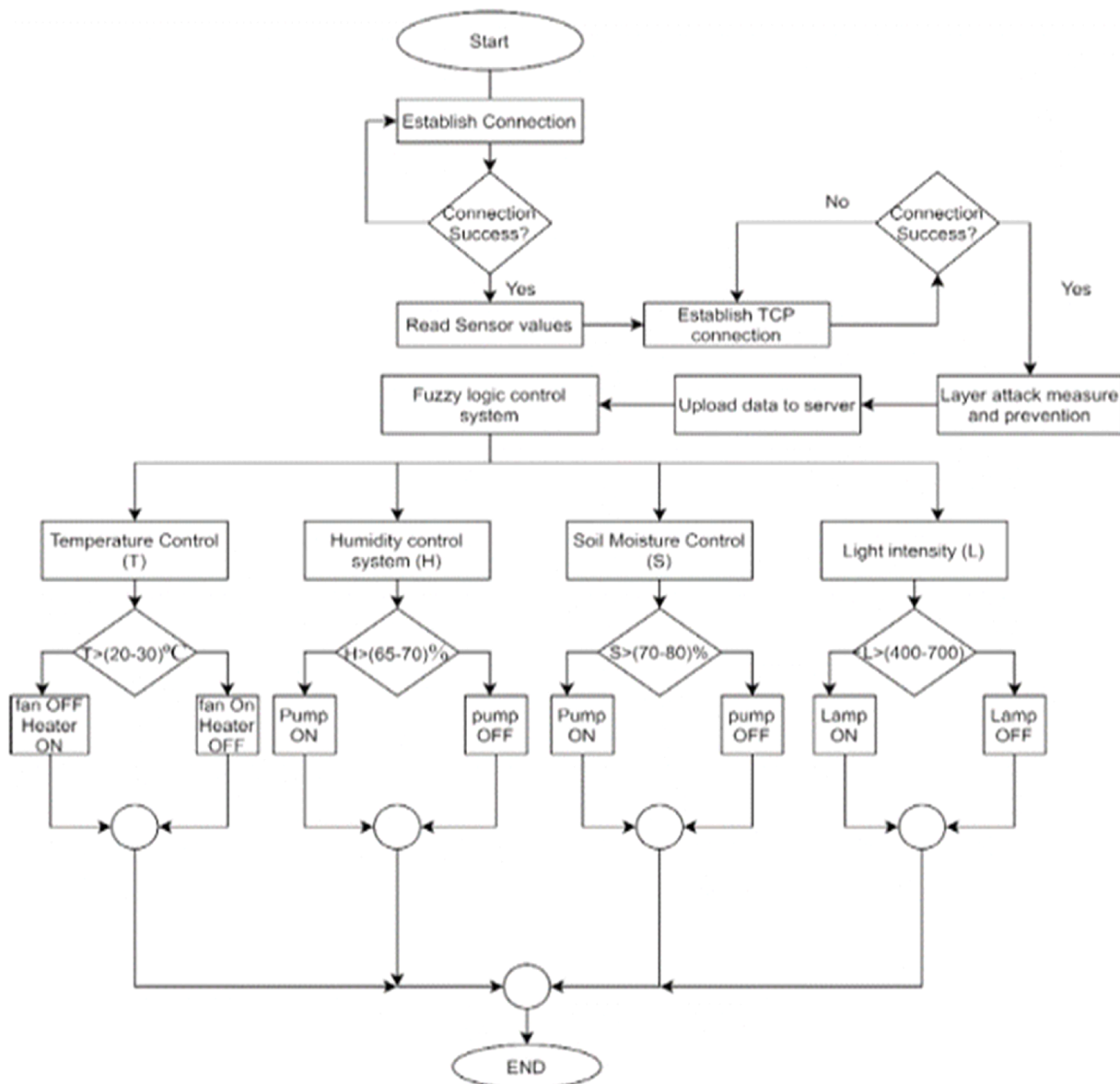


Fig. 3. Flowchart of the greenhouse monitoring and control system.

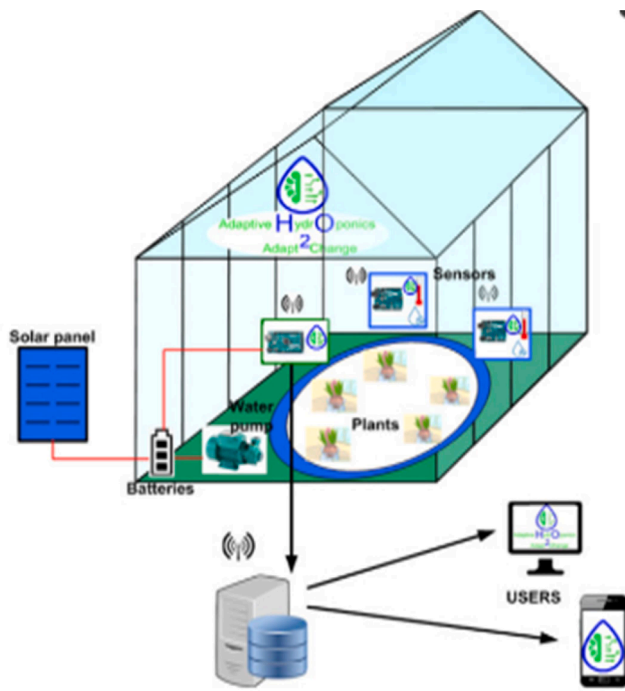


Fig. 4. Conceptual overview of the iPONICS system.

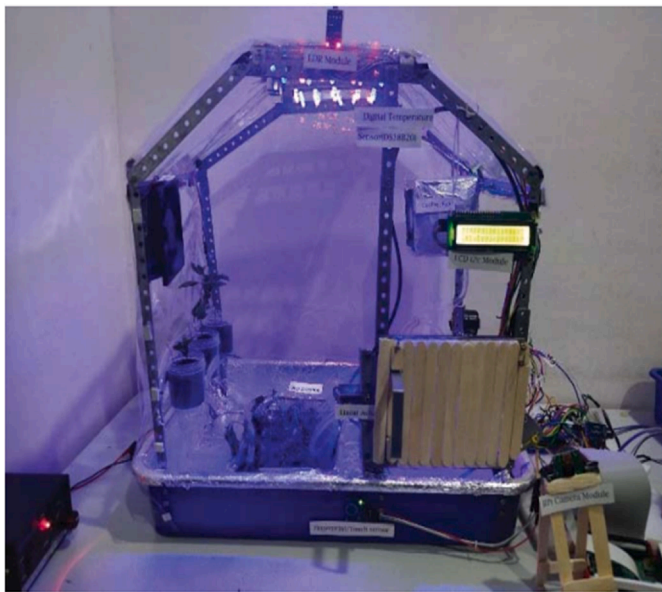


Fig. 5. Model of the greenhouse system.

conditions remotely. Its key advantages include reduced energy and water consumption, the creation of autonomous systems with minimal resource use, and improved agricultural productivity and food sustainability [34,35]. A smart greenhouse typically consists of three main components: microcontrollers, actuators, and sensors. Sensors collect critical data on soil, plant, and climatic conditions to support data-driven decisions, while actuators manage various control mechanisms within the system. The data collection system, comprising both conventional and IoT-enabled components, facilitates data transmission through communication networks [36]. IoT-enabled systems monitor indoor environmental parameters, enabling informed decision-making and early detection of abnormal conditions. By combining sensors, communication technologies, and data analytics, Farooq et al. [37]

demonstrated that IoT enhances smart farming while addressing challenges such as interoperability and data security. Furthermore, Farooq et al. [38] developed an IoT-based smart greenhouse framework for sustainable agriculture that employs intelligent decision-making, adaptive control strategies, and real-time monitoring to regulate soil parameters, humidity, and temperature. Building on this, Farooq et al. [39] conducted a comprehensive review of IoT technologies, applications, and communication protocols in greenhouse agriculture, covering sensor networks, cloud-based data processing, and connectivity standards like Wi-Fi, LoRa (long range), and Zigbee. The study also highlighted future research opportunities involving edge computing, energy-efficient communication, and AI to enhance greenhouse management. Similarly, Hosny et al. [40] examined IoT systems in greenhouse farming, emphasizing data-driven decision-making, interoperability, and 5 G integration for developing intelligent, scalable, and secure infrastructures. In addition, Maraveas et al. [41] demonstrated the practical applications of IoT in optimizing greenhouse environmental control and resource management by integrating sensor networks, cloud computing, and intelligent control systems. Their findings highlighted IoT's potential to improve sustainability and precision in greenhouse operations, with notable gains in crop yield, water conservation, and energy efficiency.

Sofwan et al. [42] described a smart greenhouse employing IoT technology to monitor environmental parameters, such as humidity and temperature, for optimizing plant growth. Using the Message Queuing Telemetry Transport (MQTT) protocol, data were transmitted efficiently, as shown in Fig. 2, with average measurement errors of 7.22 % and 0.9 °C. However, the system exhibited certain limitations, including limited modularity, high cost, interoperability issues, reduced autonomy, and structural complexity.

Soheli et al. [43] developed an IoT-based intelligent greenhouse monitoring system that uses sensors and a WSN to control environmental parameters such as temperature, humidity, and soil moisture. Fig. 3 illustrates the flowchart of this study. The system enhances prediction accuracy and learning efficiency; however, it does not examine its alignment with real-world greenhouse dynamics.

Tatas et al. [44] developed iPONICS, a low-cost IoT-based control and monitoring system for hydroponic greenhouses, as illustrated in Fig. 4. The system monitors environmental parameters to ensure optimal crop growth and is designed for off-grid operation. It incorporates a fuzzy inference engine (FIE) to determine irrigation duration. The system was tested in an experimental greenhouse and operated successfully in a laboratory setting for over a year. However, the study reported certain limitations, including the absence of long-term reliability data and limited precision in nutrient prediction models.

Kirci et al. [45] developed an intelligent greenhouse prototype to monitor plant growth and compare it with naturally cultivated plants. The greenhouse environment was controlled using an Arduino microcontroller. Their experimental results showed that plants grown in the smart greenhouse showed better growth performance and used 16.5 % less water than those cultivated in conventional pots. However, high humidity levels led to insect infestations.

Ragaveena et al. [46] designed an automated greenhouse system integrating ML and IoT technologies to monitor and regulate environmental parameters for optimal plant growth (Fig. 5). The system employs ZigBee technology for real-time parameter measurement and communication, along with a mobile application for generating alerts. Real-time data are displayed on the ThingSpeak cloud platform, enabling efficient management of temperature, humidity, and soil moisture. The system reduces reliance on manual labor and minimizes crop failure. Future scope may focus on analyzing sensor interrelationships to enhance cost efficiency and enable fruit growth detection.

Budiyanto et al. [47] designed and implemented circuits integrating Wi-Fi technology to collect sensing data from WSNs and transmit it to online stations. The system supports applications, such as greenhouse monitoring and remote agriculture, by collecting and displaying data on

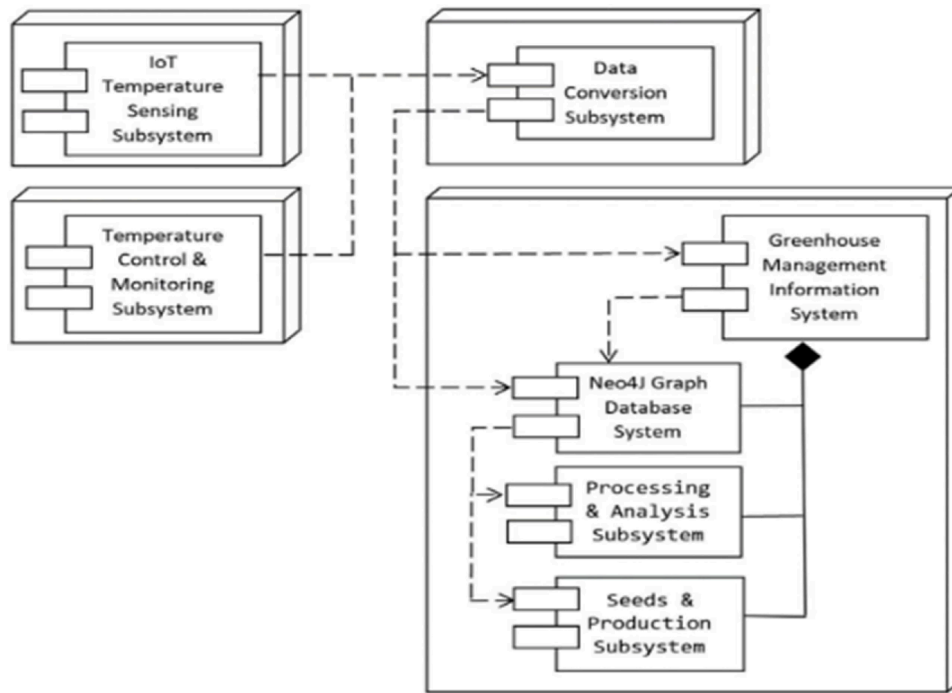


Fig. 6. Architectural design of the greenhouse system.

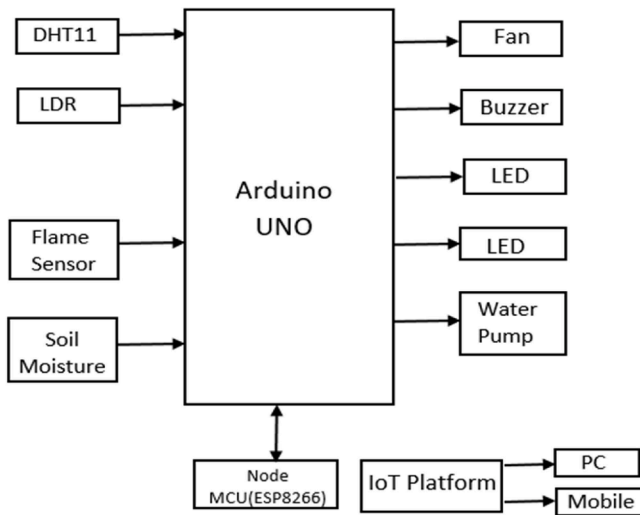


Fig. 7. Block diagram of the IoT-based greenhouse monitoring and control system.

environmental parameters including humidity, temperature, light, and movement. However, powering the hardware was challenging because of high current consumption.

Subahi and Bouazza [48] developed an energy-efficient system designed to manage large volumes of IoT data. A Petri Net model was employed to generate reference temperatures for the temperature regulating block. Fig. 6 illustrates the architecture of the greenhouse monitoring and controlling system. Simulation results showed a 38.45 % reduction in energy consumption compared to conventional controllers. However, the study's use of reference temperatures and solar ray angles was limited as these parameters were illustrative and may require adjustment for specific greenhouse conditions.

Sujin et al. [49] developed a system that automates the monitoring and control of greenhouse parameters using an Arduino microcontroller and various sensors. As illustrated in Fig. 7, the system connects sensors

and controls actuators such as fans, water pumps, and artificial lights. It can be monitored remotely through an IoT platform. However, the study identified certain limitations, including high energy consumption and the need for integration with renewable energy sources, additional sensors, and advanced irrigation techniques.

Widiyati et al. [50] developed an IoT-based greenhouse monitoring system designed to collect real-time microclimate data. Despite limitations—such as operation within a local network and the absence of automated controls—the system achieved a 99.76 % data collection success rate during a seven-day testing period. The findings indicate that the system can enhance greenhouse management by providing accurate and timely data for informed decision-making. However, the prototype-level system's validation under real greenhouse conditions remains limited owing to its reliance on basic environmental sensors and lack of integration with automated control mechanisms.

Yang [51] developed an intelligent agriculture service system using a LoRa-based WSN. The system employs sensors to monitor soil and environmental parameters, thereby enabling automated control of lighting, irrigation, and cooling. A user-access platform, available on both computers and mobile devices, allows users to analyze data, monitor crop growth, and manage equipment. The system was successfully implemented in a dragon fruit greenhouse, indicating its practicality and suitability for remote areas with limited cellular data coverage. However, in greenhouse monitoring systems, the reliability of data transmission may be affected by the limited range and potential signal interference associated with traditional wireless sensor technologies.

Andrianto et al. [52] developed and evaluated an IoT-based smart greenhouse system for hydroponic farming, aiming to enhance agricultural production without the use of pesticides. The system monitored internal parameters and controlled actuators through a smartphone application, as shown in Fig. 8. Future research will focus on developing a service computing platform to integrate additional functionalities into the smart greenhouse system. However, an extended testing period is required to strengthen the system's reliability.

Sun et al. [53] developed a low-power wide-area network (LPWAN) information monitoring system integrating narrowband IoT (NB-IoT) and LoRa technologies. The system showed a low communication packet

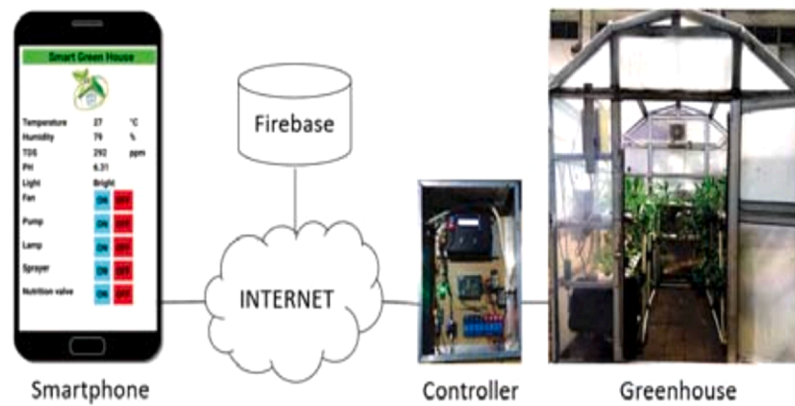


Fig. 8. Architecture of the smart greenhouse system for hydroponic farming.

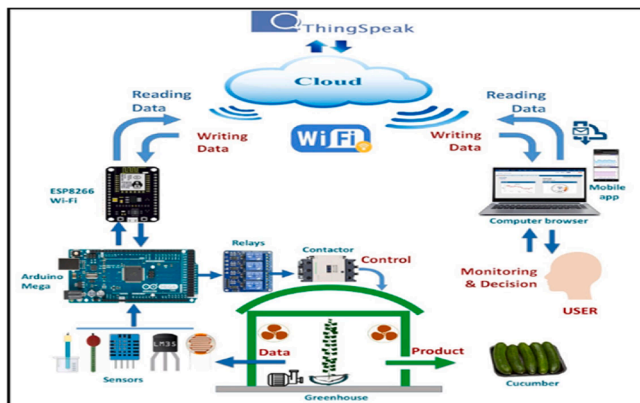


Fig. 9. Architecture of the cloud-based IoT system for greenhouse management.

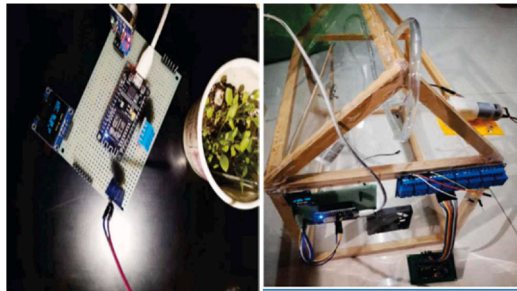


Fig. 10. Prototype of the integrated greenhouse system.

loss rate and effective data transmission over distances of up to 1.6 km in complex environments. However, factors, such as antenna design, electromagnetic interference, and physical obstructions, were found to affect the radio transmission range. Future research should focus on testing communication across a greater number of LoRa nodes and improving algorithm design.

Similarly, Sagheer et al. [9] developed a cloud-based IoT infrastructure to optimize soilless cucumber cultivation in greenhouses. The platform integrates sensors, controllers, and actuators to enable long-distance monitoring and management, as shown in Fig. 9. However, high initial investment and long-term sensor maintenance may limit the system's overall effectiveness.

Wan et al. (2019) developed a greenhouse management system featuring dynamic environmental monitoring and remote control, as shown in Fig. 10. The system employs an ESP8266 NodeMCU board and

various sensors to monitor environmental parameters. It is cost-effective, suitable for small greenhouses and indoor gardens, and applicable to contemporary smart agriculture.

Araújo et al. [54] developed a cost-effective, compact, and energy-efficient system for monitoring greenhouse parameters. The system records microclimate data on the IBM Watson Cloud platform and alerts staff to deviations. Its hardware is based on a TI CC2650 Sensor Tag, while the software integrates a smartphone application and the IBM Cloud Platform with Node-RED for data processing. The system provides a safe environment for plants and incorporates an ML model for plant health classification. Although the proposed IoT-based device was designed to be compact, cost-effective, and accurate for greenhouse weather monitoring, its effectiveness and adoption in real-world applications would benefit from stronger integration with existing agricultural practices, given its current size limitations.

Harisha et al. [55] presented an AI-controlled smart IoT-based greenhouse monitoring system for precision agriculture (Fig. 11). The system optimizes environmental parameters within greenhouses to enhance crop yields and resource efficiency. It integrates sensors, Wi-Fi control, and a real-time cloud database, with AI automatically managing temperature, moisture, and lighting automatically. The system's stability, robustness, and reliability were validated through comparative studies with commercial and research greenhouse experiments.

Raj et al. [56] developed an IoT-based system for greenhouse environments. The system employs sensors, a Netduino 3 Wi-Fi microcontroller, and cloud services to monitor environmental parameters and control the greenhouse climate. Data are stored in a database and transmitted to farmers for remote monitoring. A motor driver circuit is used for temperature regulation, as shown in Fig. 12. However, the system requires frequent human inspections, which may interrupt staff workflow and reduce operational efficiency, potentially affecting crop yields. Additionally, temperature and humidity sensors need routine maintenance to prevent productivity losses.

Ibrahim et al. [57] proposed and implemented a tiered IoT architecture (Fig. 13) for remote greenhouse monitoring and control. The system showed high accuracy in maintaining optimal environmental conditions and enabling real-time data collection, thereby enhancing crop yield, resource efficiency, and automation in smart agriculture.

To optimize agricultural water use, Kamienski et al. [58] developed a smart water management platform that applies IoT-based precision irrigation. The system monitors soil and environmental parameters, minimizing water waste and maintaining crop productivity through the integration of sensors, communication networks, cloud computing, and decision-support tools. However, the study's scalability and performance were evaluated using only a few crop types and specific experimental conditions.

By integrating sensors, communication technology, and data analytics, Ray [59] presented a comprehensive analysis of how IoT can

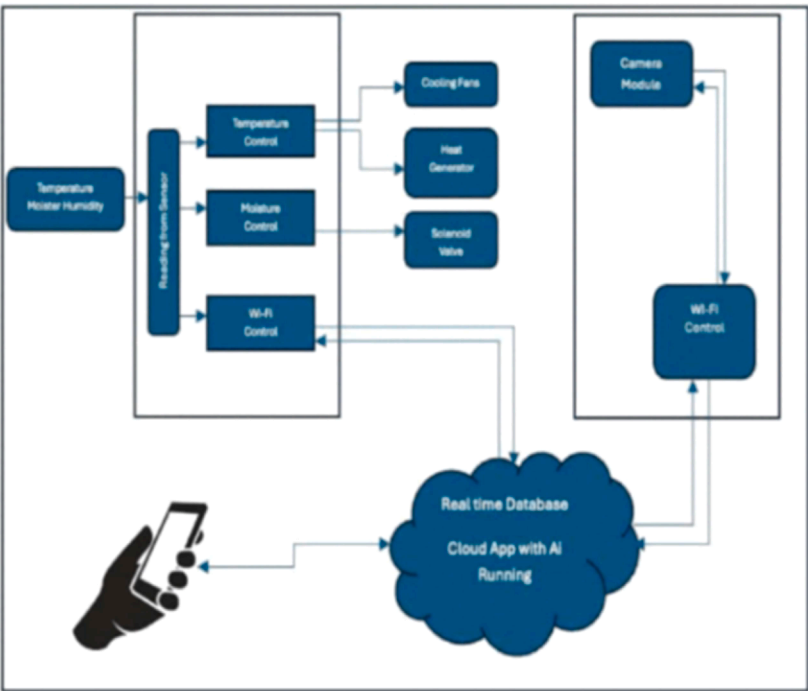


Fig. 11. Architecture diagram of the proposed AI-controlled smart greenhouse module.

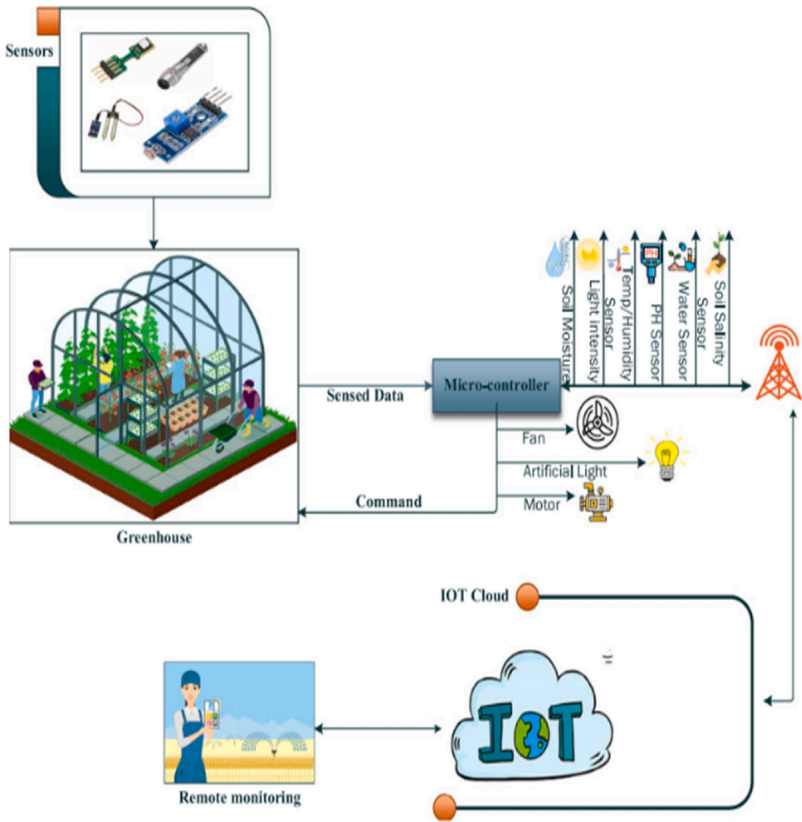


Fig. 12. Block diagram of the IoT-based greenhouse monitoring and control system.

transform smart agriculture. The study emphasized the role of IoT in enabling real-time monitoring of key agricultural parameters. By interconnecting devices through WSN and cloud-based platforms, farmers can automate irrigation, pest control, and resource management, leading to more precise and efficient farming practices. However, the

proposed method needs to be tested in real agricultural environments to assess their performance, scalability, and cost-effectiveness. This limitation arises because the study primarily focused on conceptual frameworks and theoretical discussions rather than experimental validation or field implementation.

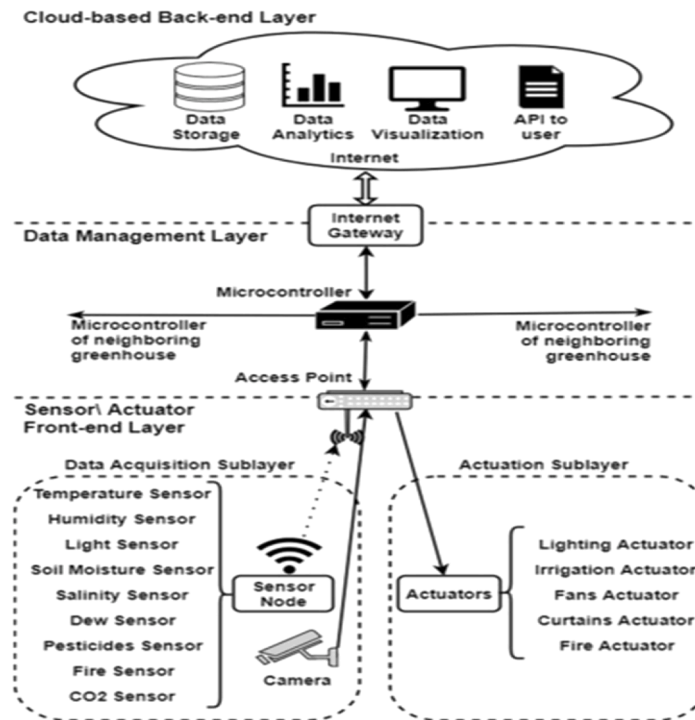


Fig. 13. Tiered IoT system architecture for greenhouse monitoring.

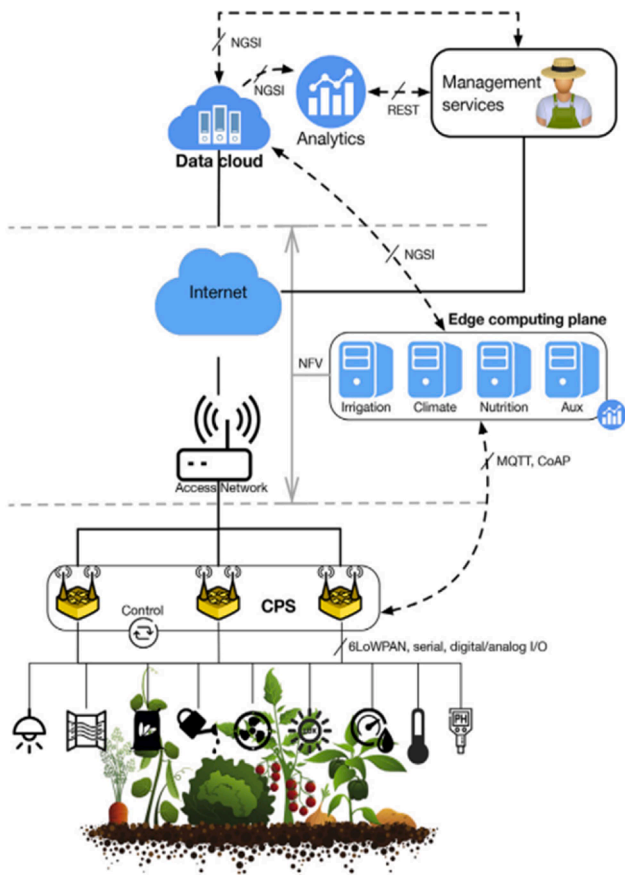


Fig. 14. Overview of the IoT system integrated with cloud computing.

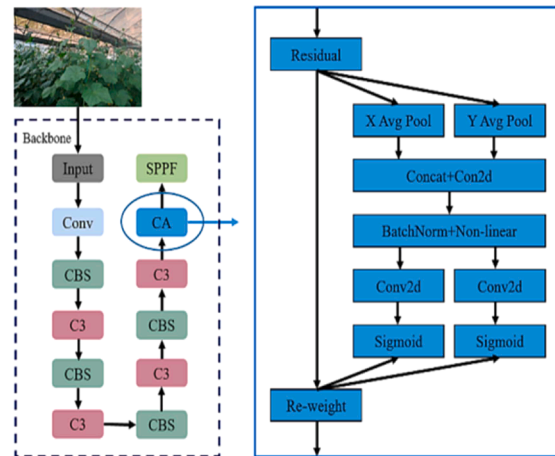


Fig. 15. Backbone structure of the YOLOv5 model with the Coordinate Attention (CA) mechanism.

Zamora-Izquierdo et al. [60] developed an IoT-based smart farming platform that integrates edge and cloud computing for real-time greenhouse monitoring and control (Fig. 14). The system employs a three-layer architecture local, edge, and cloud using MQTT and CoAP protocols to enhance data management, scalability, and reliability. When evaluated in hydroponic systems, the platform showed efficiency and resilience, although the testing conditions and quantitative evaluation were limited. However, its scalability and dependability in larger or more diverse agricultural environments may be constrained because of the study's focus on controlled greenhouse conditions.

4. Deep/machine learning techniques for plant health detection

Artificial intelligence has advanced rapidly across various domains, especially in agriculture and greenhouse management [61]. Machine learning algorithms are employed to regulate greenhouse microclimates

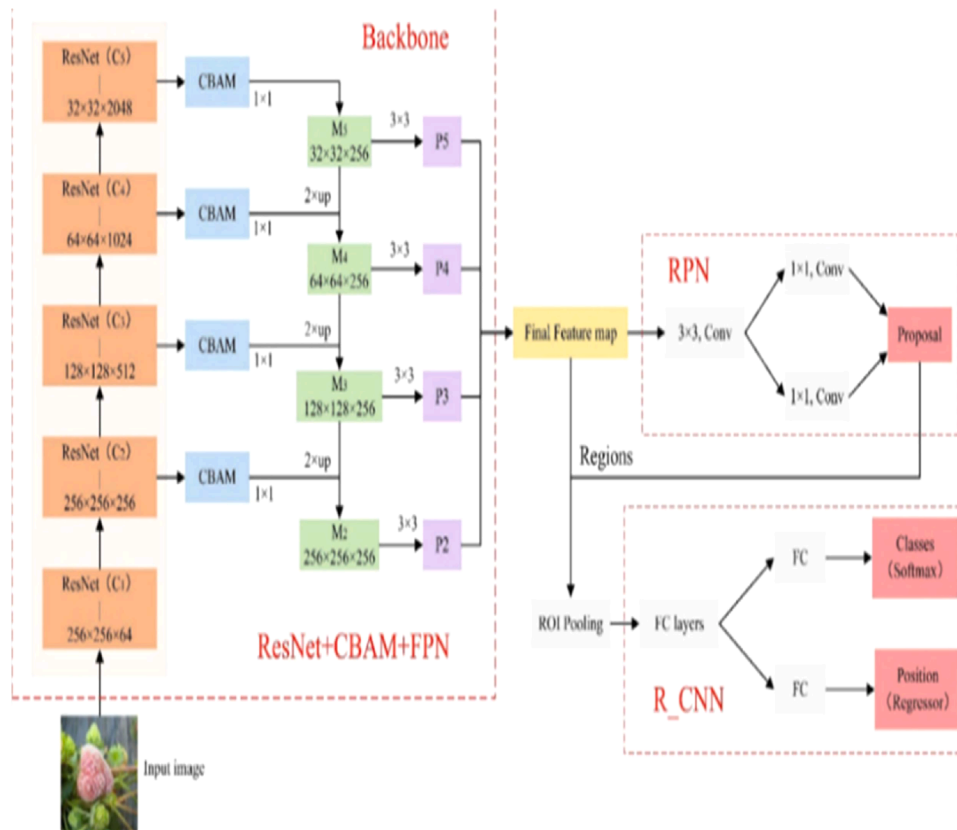


Fig. 16. Structure of the multiscale feature fusion model.

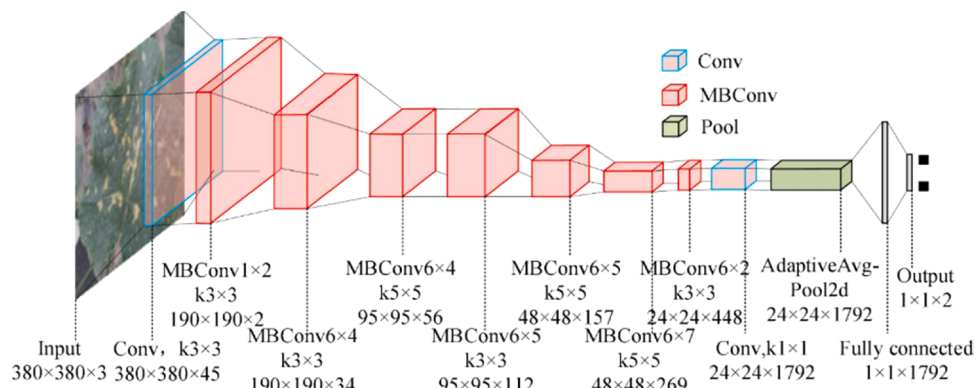


Fig. 17. Network structure of the EfficientNet-B4 model.

and predict internal environmental factors that influence crop growth and energy consumption [62]. AI-based systems can achieve performance comparable to skilled human growers while improving crop quality. Convolutional neural network (CNN) models, such as LeNet, AlexNet, VGGNet, MobileNet, Inception V3, EfficientNet, GoogLeNet, and ResNet, form the foundation of many AI applications [63]. The integration of ML and IoT technologies supports greenhouse farmers in overcoming production challenges and improving crop quality and yield [64]. Deep learning techniques have been widely applied to optimize environmental control, monitor plant growth, and detect plant diseases in greenhouses. According to Ojo and Zahid [65], DL model applications have gained significant attention in recent years; however, identifying the most suitable DL model remains a challenge.

Maraveas [66] examined recent advancements in applying AI technologies into smart greenhouse systems aimed at enhancing agricultural productivity, resource efficiency, and sustainability. The study

highlights the expanding use of AI techniques in energy management, fertigation, irrigation, climate control, pest and disease detection, and optimization through ML and DL algorithms.

Chen et al. [67] developed an improved YOLOv5 DL model for image recognition to enhance detection accuracy in cucumber canopy vine-top images. The model integrates a Coordinate Attention (CA) module and a Focal Loss function to improve recognition of small leaf targets (Fig. 15). The proposed YOLOv5-CA-LS model achieved an average recognition accuracy of 97.1 % for dense and mixed cucumber canopies, showing reliable performance under various viewing angles and lighting conditions.

Afonso et al. [68] employed the MaskR-CNN algorithm with RealSense cameras to detect and count tomato fruits in greenhouse environments. The system aimed to automate manual tasks such as phenotyping, yield prediction, and harvesting. Among the tested models, the ResNext101 architecture achieved the best performance,

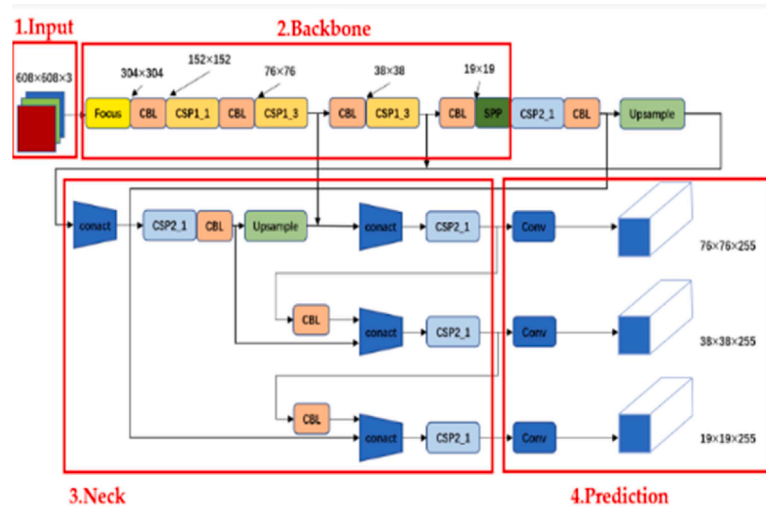


Fig. 18. Network structure of the improved YOLOv5 model for cucumber fruit recognition.

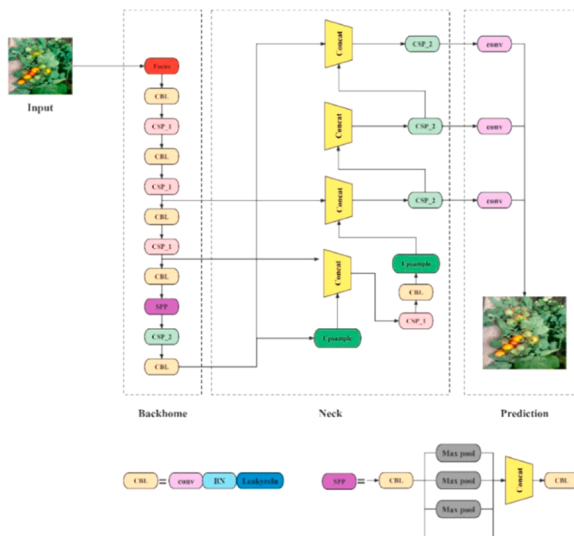


Fig. 19. Structural diagram of the YOLOv5 network model.

surpassing traditional methods. The model effectively differentiated between ripe and unripe fruits, although some color ambiguity persisted. Future research should focus on fruit detection across multiple images and incorporating depth information for improved accuracy.

Xu et al. [69] proposed the Pruned-YOLO v5s+Shuffle (PYSS) model, a real-time object detection approach for identifying melon leaf diseases in greenhouse environments. The model incorporates the ShuffleNet v2 inverted residual block for feature extraction and applies channel pruning to reduce size. When deployed on a Jetson Nano device, PYSS achieved mean average precisions (mAP) of 93.2 % and 98.2 % for melon powdery mildew and healthy leaves, respectively, representing a 6.2 % improvement in mAP and a 6.4 % increase in precision. However, the study noted that the powdery mildew spots were too small for effective feature extraction and learning, posing a challenge for detection accuracy. Future research will focus on expanding the model to identify additional melon diseases and apply it to other crops.

Zhao et al. [70] developed a multiscale feature fusion architecture based on the Faster R-CNN for accurate disease detection on strawberry leaves, flowers, and fruits, as shown in Fig. 16. The model integrates feature pyramids to enhance feature extraction and employs attention mechanisms to highlight key disease regions. It achieved an mAP of 92.18 %, outperforming other comparative models.

Zhang et al. [71] developed an EfficientNet-based approach for detecting cucumber diseases in greenhouses (Fig. 17). The model classified four leaf conditions—powdery mildew, downy mildew, combined infection, and healthy leaves. The EfficientNet-B4 model achieved 97 % classification accuracy, which was further optimized using the Ranger optimizer, yielding 96 % accuracy. The improved method showed superior performance in distinguishing visually similar cucumber diseases.

Shukla et al. [72] developed an IoT-based system for predicting grape diseases using CNN and farm monitoring. The model analyzes leaf images, collects real-time data, and classifies diseases into leaf blight, Esca-black measles, and black rot. It also provides fertilizer recommendations based on the predicted disease index and pruning date.

Similarly, Nan et al. [73] proposed WGB-YOLO, an improved YOLOv3 network for pitaya fruit detection. The model enhances detection accuracy while maintaining high speed, particularly for occluded and partially visible fruits. WGB-YOLO achieved an mAP of 86.0 %, outperforming other models and improving feature extraction capacity. However, its higher weight and parameter size compared to YOLOv7 remain notable limitations.

Wang et al. [74] improved cucumber fruit recognition in greenhouse environments using an improved YOLOv5 model that incorporates color space features (Fig. 18). An extended RGB image dataset was constructed, and twelve channels across four color spaces were evaluated using the ReliefF method. The Cr channel from the YCbCr color space was found to have the highest weight in distinguishing cucumber fruits from the background. The improved YOLOv5s model achieved higher recognition precision and accuracy compared to the standard division.

Li et al. [75] introduced YOLOv5s-tomato, an improved YOLOv5-based model (Fig. 19) for recognizing four tomato maturity stages. The model outperformed other techniques, achieving high recall, precision, and F1 score, with an mAP of 96.40 %. The improved model showed a smaller size, faster detection speed, and higher recognition precision.

Liu et al. [76] introduced YOLO-Tomato, an improved tomato detection model based on YOLOv3, designed to address complex environmental environments (Fig. 20). The model replaces rectangular bounding boxes with circular ones and integrates a dense architecture. YOLO-Tomato outperformed state-of-the-art detection methods, achieving high recall, precision, and F1 score, with an mAP of 96.40 %. However, challenges remain in accurately detecting tomatoes at different growth stages and under varying environmental conditions, highlighting the need for continued research and model optimization.

Liu et al. [77] developed a Long Short-Term Memory (LSTM) neural network to predict cucumber downy mildew in solar greenhouses

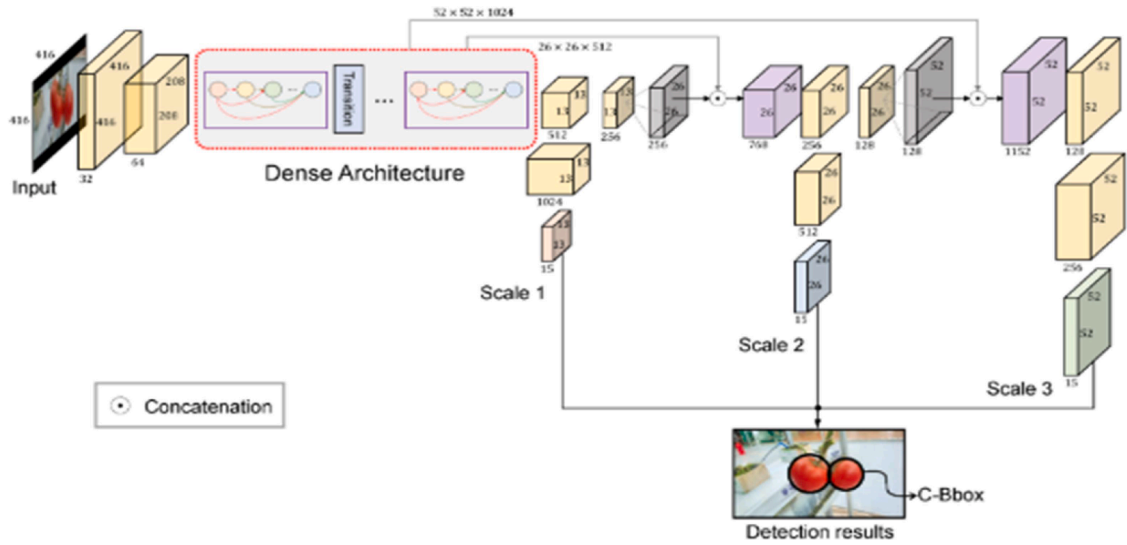


Fig. 20. Overview of the proposed YOLO-Tomato model.

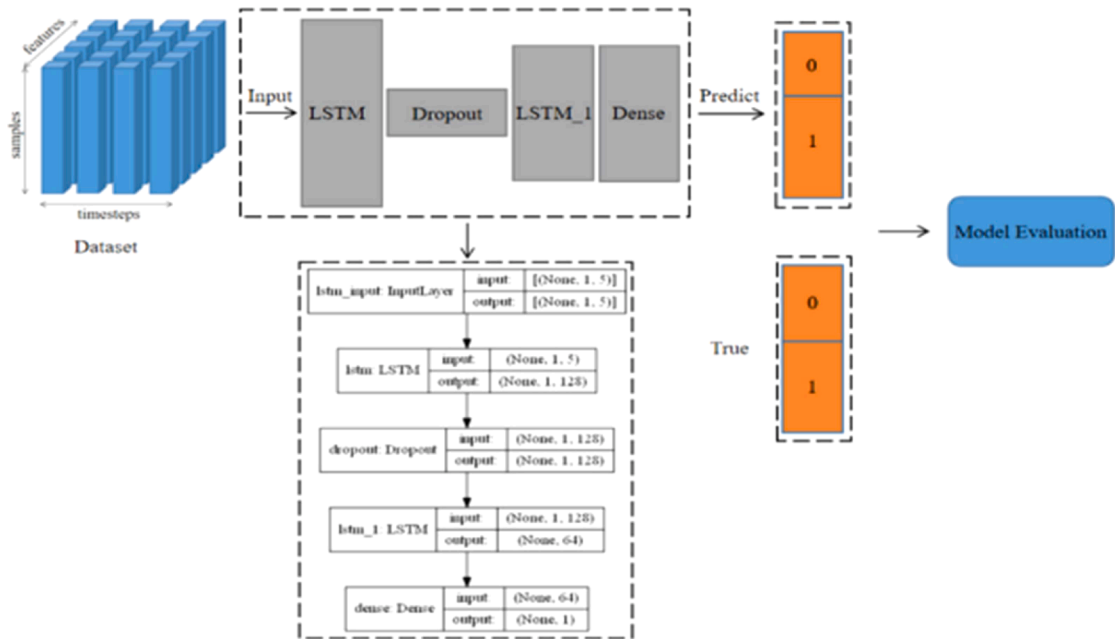


Fig. 21. Structure of the disease prediction model.

(Fig. 21). Implemented using the Keras DL framework, the model achieved a training accuracy of 95.99 % and a prediction accuracy of 90 %, outperforming the K-Nearest Neighbors and artificial neural network (ANN) models. The study highlights the potential of the LSTM model as an effective tool for real-time forecasting of cucumber downy mildew.

Badji et al. [78] applied AI to predict greenhouse air temperature using Nonlinear AutoRegressive with eXogenous input (NARX) model (Fig. 19). Trained on historical data, the model accurately predicted future air temperature values, with low root mean squared errors (RMSEs) indicating strong agreement with actual measurements. The study suggests that NARX models are effective for real-world greenhouse applications, though incorporating additional ML algorithms could further enhance performance.

Mao et al. [79] developed an automatic cucumber recognition method using DL and multi-feature fusion for harvesting robots operating in natural environments. The approach combines a multi-path CNN, a support vector machine, and color component selection to

detect cucumber regions (Fig. 22). The model achieved over 90 % in classifying cucumber image pixels, with misidentification rates below 22 %.

Vijayakanthan et al. [80] developed a deep transfer learning framework for precise classification of tea leaf diseases using six pre-trained CNN models (Fig. 23). Among these, the Xception model achieved the highest classification accuracy of 99.58 %. However, the model faced challenges in differentiating visually similar symptoms between diseases such as algal leaf spot and grey blight, leading to occasional misclassifications. Future work should focus on developing lightweight models suitable for deployment on embedded systems.

Proietti et al. [81] employed edge intelligence and DL to automate greenhouse operations and detect anomalies in plant growth and equipment performance. The system utilizes an LSTM Encoder–Decoder architecture that learns “normal” patterns from sensor data to identify deviations in greenhouse operations and suggest corrective actions.

Karar et al. [82] developed a mobile application based on the Faster

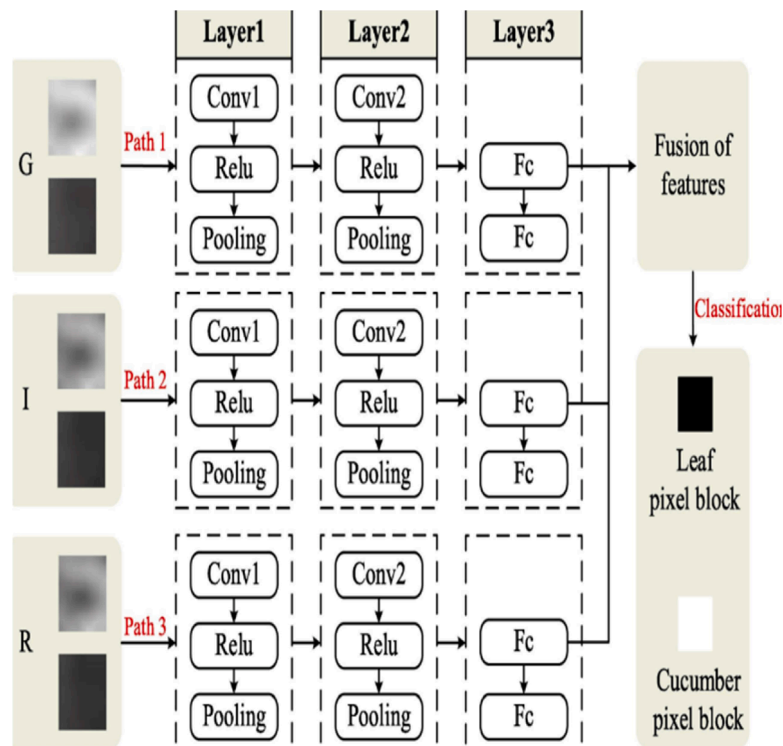


Fig. 22. Network structure of the multi-path CNN model.

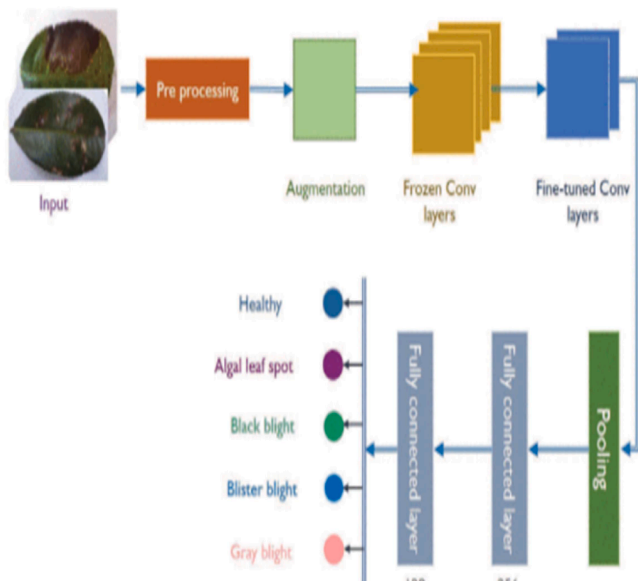


Fig. 23. Proposed architecture of the convolutional neural network (CNN).

R-CNN DL technique for image-based agricultural pest recognition. The study highlights the economic impact of pest infestations and the importance of advanced solutions to reduce pesticide use and boost crop yields. The proposed Faster R-CNN model achieved the highest recognition accuracy of 99.0 % across all tested pest images. To support farmers, the application integrates a database of recommended pesticides linked to identified pests. Although specific performance metrics were not provided, a confusion matrix was used to assess classification performance.

Cui et al. [83] investigated the effects of aerated irrigation and varying nitrogen (N) application rates on cucumber yield and nitrogen use efficiency in greenhouse settings. Compared to traditional

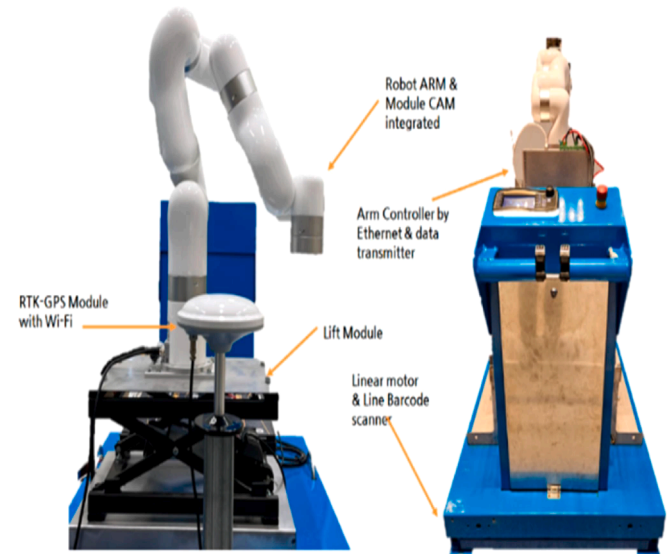


Fig. 24. Structure of the image acquisition device.

approaches, aerated irrigation combined with optimized N rates significantly increased cucumber yield and enhanced nitrogen use efficiency. These improvements were achieved using an ANN model to optimize irrigation and fertilization. The findings suggest that integrating AI models such as ANN can enhance crop productivity, reduce nutrient losses, and support sustainable greenhouse cucumber cultivation.

Abdulridha et al. [84] evaluated the detection of powdery mildew in squash at different disease stages using unmanned aerial vehicle (UAV)-based hyperspectral imaging in conjunction with artificial intelligence. A CNN model was employed to analyze the hyperspectral data. The findings indicate that combining AI models with UAV imaging enables early and accurate disease detection, supporting timely interventions and improved crop management in both field and



Fig. 25. SWEEPER robot designed for automated sweet pepper harvesting in greenhouse environments.

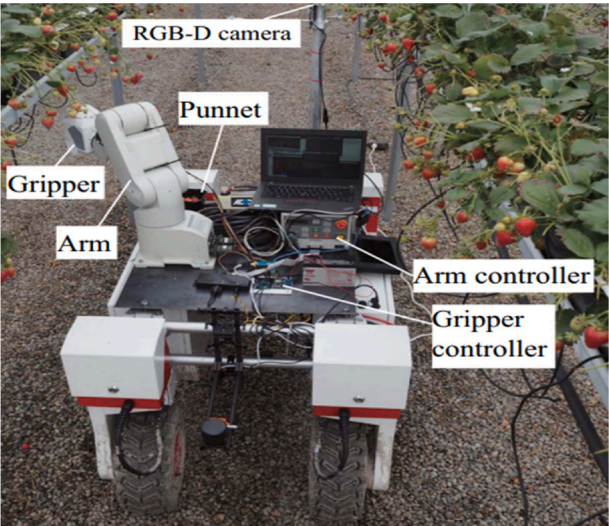


Fig. 26. Autonomous strawberry-picking robot.

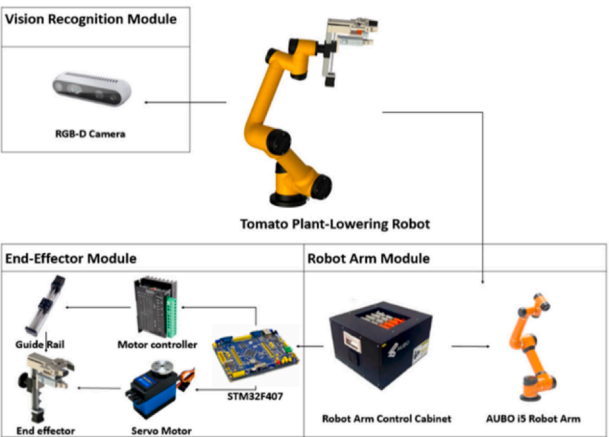


Fig. 27. Tomato lowering robot system.

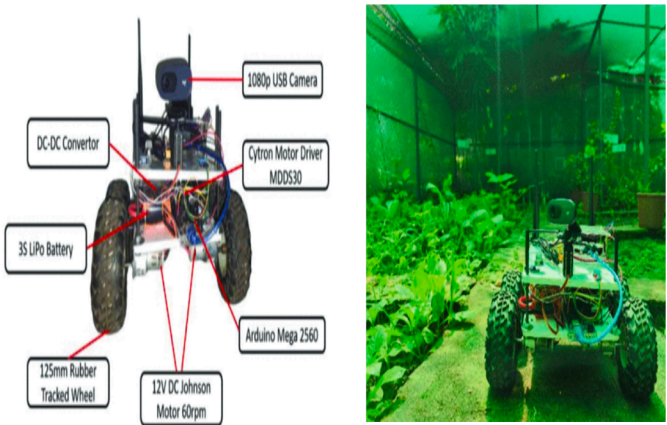


Fig. 28. Agrobot operating in a greenhouse environment.



Fig. 29. Design of the robotic system for tomato leaf disease detection.

greenhouse environments.

5. Robotics for automation

Robotics has become an essential field across multiple sectors, including agriculture [85]. Initially applied to harvesting, robotic systems are now used for planting, spraying, and crop monitoring. These technologies enhance greenhouse sustainability through automation and data-driven management, enabling early disease detection and quality assessment of produce. Real-time monitoring supports improved productivity and competitiveness in modern food production [86]. Despite extensive research, the practical implementation of agricultural robotics remains limited [87,88]. However, robotics holds great potential to transform urban agriculture by enhancing economic efficiency, sustainability, and productivity. Integrating robotic systems with sensors can optimize growing conditions, enabling advanced management strategies aligned with sustainable development goals (Doss et al., 2024).

Kim et al. [86] developed an AI-based environmental control system for tomato disease image classification to enhance greenhouse sustainability. The system achieved 94 % accuracy in disease classification and enables autonomous navigation within the greenhouse. It comprises a robotic arm, an RGB camera, a pan-tilt-zoom (PTZ) support, adjustable lift, light sensors, and RTK-GPS (Fig. 24). The integration of AI and automation improves operational efficiency and sustainability by reducing human intervention, optimizing greenhouse management, and supporting pest control decisions even in the absence of specialists.

Moreira et al. [89] compared conventional color space segmentation methods (HSV) with DL techniques for and class tomato detection and classification in greenhouses. The AgRob v16 robot, operated manually,

Table 1
Summary of IoT applications in greenhouse systems.

No.	Greenhouse application	Method used	Description/performance	Citation
1	Monitoring	• MQTT • Cloud computing	Developed a system that utilizes microclimate sensors to support optimal plant growth.	[42]
2	Monitoring and control	• MQTT • GSM	Developed a system using neuro-fuzzy sensors that enable farmers to make decisions and transmit them to the system.	[43]
4	Monitoring and control	• WSN • Zigbee • GSM/GPRS	Improved an IoT-based hydroponic system's responsiveness and sustainability, paving the way for autonomous smart agriculture setups.	[44]
5	Monitoring	Bluetooth	Developed and tested a small-scale prototype of a smart greenhouse, highlighting potential scalability challenges.	[45]
6	Monitoring and control	ZigBee	Developed a fully automated greenhouse using IoT and ML for optimal plant growth and remote monitoring.	[46]
7	Monitoring	• Wi-Fi, • Online server	Designed circuits integrating Wi-Fi technology, with scope for expanding the network using additional sensor nodes and mobile sensors.	[47]
8	Monitoring and control	5 G wireless MIMO method.	Developed system incorporating a Petri Net model to regulate and monitor greenhouse temperature, enhancing agricultural production in extreme climates.	[48]
9	Monitoring and control	Wi-Fi	Developed a smart greenhouse system for plant growth using a low-cost, efficient programmable module and NodeMCU ESP8266, controlled via a mobile device.	[49]
10	Monitoring	MQTT	Developed a system powered by an Arduino Uno microcontroller with YL-69 and DHT11 sensors, providing real-time data via a web-based interface and Wi-Fi for remote monitoring.	[50]
11	Monitoring and control	• WSN • LoRa • Bluetooth	Utilized sensors for soil and environmental detection; feasibility tested using an intelligent agriculture service platform, JAVA mobile app, and Android Studio software.	[51]
12	Monitoring and control	LoRa	Designed a system to effectively monitor and control greenhouse parameters via a smartphone application, storing temperature and humidity in a real-time database.	[52]
13	Monitoring	• NB-IoT • LoRa	Developed a hybrid system offering efficient	[53]

Table 1 (continued)

No.	Greenhouse application	Method used	Description/performance	Citation
14	Monitoring and control	Wi-Fi	long-range communication and low-power operation. Developed a multi-layered internet platform for greenhouse in arid regions using sensors, assessing its impact on crop production and quality.	[9]
15	Monitoring	Wi-Fi & HTTP	Designed a prototype system for indoor or small-sized gardens to remotely control and monitor internal environmental parameters, reducing labor costs.	[97]
16	Monitoring	Wi-Fi	Developed a weather monitoring and notification system emphasizing cost-effectiveness, compactness, and power efficiency.	[54]
17	Monitoring and control	Wi-Fi	Demonstrated how real-time data analysis and AI algorithms enable precise control over environmental factors, predict crop yields, analyze energy use, and support informed decision-making.	[55]
18	Monitoring and control	Wi-Fi	Developed an automated system that effectively manages and controls the greenhouse environment, offering a robust solution for greenhouse management through SMS alerts.	[98]
19	Monitoring and control	Wi-Fi & Bluetooth	Designed a system for greenhouse management that addresses key challenges in traditional farming affecting saffron cultivation.	[99]
20	Monitoring and control	• ZigBee • Wi-Fi	Designed an IoT-based system for real-time monitoring and control of temperature, humidity, and light intensity in greenhouses.	[57]
21	Control	• LoRa • MQTT • Cloud computing	Developed an IoT-enabled precision irrigation platform to optimize water use in controlled greenhouse environments.	[58]
22	Monitoring	• ZigBee • Cloud-based gateway	Implemented a multi-sensor wireless network for measuring CO ₂ , humidity, and temperature to support greenhouse optimization.	[59]
23	IoT-based greenhouse management platform	• Edge computing • Wi-Fi	Developed an edge-cloud integrated system enabling local decision-making for greenhouse control and data processing.	[60]

The second section (summarized in Table 2) presents the use of DL/ML techniques. Current research primarily focuses on developing models that outperform previous approaches in terms of accuracy, speed, and robustness. However,

many studies focus mainly on algorithmic enhancements without addressing real-world implementation issues.

captured 449 greenhouse images using a ZED camera connected to a Raspberry Pi Model B, maintaining a 60 % overlap between photos. The resulting dataset was used to train machine learning models for tomato detection and classification, enhancing crop management and harvesting efficiency.

Ge et al. [90] developed the YOLO-DeepSORT model, a greenhouse inspection robot that uses QR codes to count tomato flowers during movement. The system integrates an enhanced YOLO model with a target-tracking algorithm to prevent double-counting and improve yield prediction. Although the model's performance can be affected by environmental complexity and movement-induced instability, it achieves high precision, recall, F1-score, and mAP, demonstrating reliable tracking and counting accuracy.

Arad et al. [91] developed and tested SWEEPER (Fig. 25), a robotic system designed for harvesting sweet pepper fruits in greenhouses. The robot integrates motion control, grasp pose estimation, and fruit detection algorithms. The study highlighted the importance of optimizing crop conditions and selecting appropriate cultivars to enhance harvesting efficiency. When crop visibility was improved and fruit clustering reduced, the robot's harvest success rate increased to 61 %.

Xiong et al. [92] designed and evaluated an autonomous strawberry-harvesting robot for polytunnel environments (Fig. 26). The robot employs an obstacle-separation algorithm and a mobile platform that can push aside obstacles, enabling efficient harvesting of strawberries growing in clusters. The system comprises two grippers, a dual-arm manipulator, a mobile platform, and an RGB-D camera, facilitating continuous and precise strawberry harvesting along crop rows.

Zhang et al. [93] developed an automated robot for lowering tomato plants using machine vision and DL techniques to enhance crop growth and yield (Fig. 27). The robot integrates a depth camera, YOLOv5s, and RepC3 module for object detection and positioning. Results showed improved recognition accuracy, positioning precision, and operational efficiency. However, limitations included the robotic arm's inability to reach designated positions, wiring disconnections, and hook sway. Future research will focus on optimizing the vision systems, point location methods, white object recognition, and end-effector mechanical design.

Jadav et al. [94] introduced an AI-driven, four-wheeled robot for targeted plant disease surveillance in greenhouse environments (Fig. 28). The system employs the YOLOv8 model for accurate and rapid disease detection and integrates IoT sensors for continuous environmental monitoring. High accuracy, recall, and mAP values indicate the system's efficacy in identifying and segmenting affected areas. The accompanying "Krishi Sahayak" mobile application enables real-time data access, facilitating informed decision-making and improved crop management.

Mac et al. [95] developed an intelligent greenhouse robotic system for autonomous navigation and early tomato leaf disease detection (Fig. 29). The system integrates an enhanced Deep Convolutional Generative Adversarial Network (DCGAN) with a ResNet model, achieving 99.69 % accuracy using ResNet-152. The improved dataset minimizes misclassification and improves class differentiation, ensuring efficient tomato row coverage with minimal disturbance.

Milella et al. [96] developed a multi-sensor robotic platform for ground mapping and environmental estimation beyond the visible light spectrum. The system integrates RGB, thermal, and hyperspectral sensors to collect detailed information on crop and soil conditions, enabling precise identification of stressors and plant health indicators. Through automated, non-invasive, and data-rich monitoring, the robot effectively supported precision farming, showing high accuracy in mapping and assessing diverse agricultural landscapes.

Table 2

Summary of ML/DL applications in greenhouse environments.

No.	Greenhouse Application	Method/Tool used	Description or performance	Citation
1	Detection	- YOLOv5-CA-LS - Canon G7X Mark II for the imaging sensor.	Achieved an average recognition accuracy of 79.1 % for segmentation and detection of sweet pepper.	[67]
2	Detection	- R-CNN - RealSense camera	Detected tomato fruits with precision above 0.88 and recall above 0.91.	[68]
3	Detection	- Pruned-YOLOv5s + shuffle (PYSS) - Vivo IQOO Neo5 camera	Achieved an mAP@0.5 of 93 % for powdery mildew detection and 98.2 % for real melon leaves.	[69]
4	Detection	- Faster R-CNN with multiscale features (ResNet, FPN, and CBAM block) - Sony digital camera	Detected healthy strawberries and seven strawberry diseases with an mAP of 92.18 % and an average detection time of 229 ms.	[70]
5	Identification and classification	- EfficientNet-B4 - Digital camera	Achieved 97 % accuracy for cucumber leaf disease identification and classification.	[71]
6	Classification	- CNN - Android smartphone camera	Classified four major grape leaf diseases affecting grape production.	[72]
7	Multi-class fruit detection	- WGB-YOLO - Stereo cameras	Achieved 86.0 % mAP for pitaya detection—96.0 % for non-occluded fruits, 84.4 % for clustered or crossed fruits, and 77.6 % for occluded fruits.	[73]
8	Object detection and recognition	Enhanced YOLOv5s	Achieved recognition precision of YOLOv5s 85.19 %; AP, F1, recall rate, and mAP improved by 8.03 %, 7 %, 8.7 %, and 8 %, respectively.	[74]
9	Recognizing	Improved YOLOv5	Achieved 95.58 % precision and 97.42 % mAP—enhancements of 0.11 % and 0.66 %, respectively.	[75]
10	Tomato detection	YOLOv3	YOLO-Tomato achieved a correct identification of rate of 94.58 %.	[76]
11	Disease prediction	Long Short-Term Memory (LSTM) neural network	Predicted cucumber downy mildew with 90 % accuracy, 94 % precision, 89 % recall, and a 91 % F1-score.	[77]
12	Prediction	Nonlinear Autoregressive with exogenous input (NARX) model	Accurately predicted greenhouse air temperature with RMSE values ranging from 1.74 % to 5.81 %, 1.40 % to 3.20 %, and 0.992 to 0.996.	[78]
13	Recognition	(I-RELIEF)/SVM feature classification. - Sony IMX386 sensor	Combined MPCNN and SVM approach achieved cucumber recognition accuracy rate above 90 %.	[79]
14	Classification	Pre-trained CNN	Achieved 99.58 % classification accuracy	[80]

(continued on next page)

Table 2 (continued)

No.	Greenhouse Application	Method/Tool used	Description or performance	Citation
15	Anomaly detection	• LSTM Encoder-Decoder architecture • Sony IMX219 8-MP sensor	using 5-fold cross-validation. Utilized a single LSTM layer (64 units) for anomaly detection in plant growth and control equipment.	[81]
16	Pests recognition	Faster R-CNN	Achieved 99.0 % accuracy for pest recognition with real-time identification; outperformed SSD MobileNet (86 %) and BP neural networks (50 % accuracy).	[82]
17	Irrigation and fertilization control	Artificial Neural Network (ANN)	Improved cucumber yield by 17.5 % and nitrogen productivity by 29 %	[83]
18	Disease detection	Convolutional Neural Network (CNN)	Achieved 82 % accuracy for asymptomatic and 99 % for late-stage powdery mildew detection	[84]

The third section (Table 3) explores the use of robotics in greenhouse operations. These studies generally concentrate on improving environmental management, automating harvesting, supporting plant growth, and detecting diseases under controlled conditions. A notable finding is the limited research on the use of robotic systems specifically designed for disease detection or plant health monitoring. Overall, the integrated application of IoT, ML/DL, and robotics remains underexplored. This lack of synergy represents a significant research gap and an opportunity to develop fully automated, intelligent greenhouse systems that can sense, analyze, and respond to environmental challenges in real time.

6. CRitical challenges faced by advanced technology in smart greenhouse operations

This assessment reveals that while emerging technologies show great potential for various greenhouse applications, several obstacles hinder their successful real-world implementation.

The practical adoption and scalability of IoT, ML/DL, and robotics in greenhouse agriculture face multiple interconnected challenges. One major limitation is the scarcity of large, publicly available annotated datasets; most studies rely on small, proprietary image collections focused primarily on a few crops—such as cucumbers, tomatoes, and strawberries. Consequently, these models exhibit limited generalizability across diverse plant species, growth stages, and varying environmental conditions, including lighting and humidity. Additionally, inconsistent evaluation metrics across studies make direct performance comparison difficult, leading to varying and often incomparable results. Many models are still developed and tested under controlled environments with limited validation in real-world greenhouse conditions. These issues are further intensified by the energy and computing constraints of robotic and embedded IoT systems, which restrict the deployment of advanced simulations and DL models in field conditions. Moreover, most existing systems depend solely on RGB imagery, while sensor fusion approaches remain underdeveloped despite their potential to integrate multi-modal sensor data such as thermal, hyperspectral, and depth information. Future research should therefore focus on creating standardized datasets, developing lightweight and effective ML/DL models, and establishing robust multi-sensor fusion frameworks suitable for real-time applications in resource-limited greenhouse environments.

Table 3

Summary of robotics use in greenhouses.

No.	Greenhouse application	Method used	Description or performance	Citation
1	Harvesting	AgRob v16 robot traversed greenhouse inter-rows and used a ZED camera to capture RGB images of tomato plants.	YOLOv4 achieved an F1-score of 85.81 %, outperforming SSD MobileNet v2 in tomato detection.	[89]
2	Detect and count	A camera-mounted inspection robot collected plant growth data.	Achieved 99.5 % precision, recall 98.4 % recall, 98.9 % F1-score, and 95.8 % mAP in detecting and counting plant growth stages.	[90]
3	Harvesting	The SWEEPER prototype included a Fanuc LR Mate 200iD robot arm, an automated cart with a scissor lift, an end-effector with cutting mechanism, an imaging system, and a fruit-catching device.	Achieved a 61 % harvest success rate under optimized crop conditions and 18 % under standard conditions.	[91]
4	Harvesting	The robot utilized an Intel R200 RGB-D depth camera, a single-rail dual-arm manipulator with two grippers, the Thorvald platform, and Hokuyo LIDAR for navigation.	Achieved a 50 %–97.1 % first-attempt success rate for harvesting isolated or partially surrounding strawberries. The pick success rate increased to 75 %–100 % after a second attempt.	[92]
5	Disease detection	A 4 × 4 wheeled robot equipped with a Cytron MDDS30 motor driver.	Successfully trained the YOLOv8 model on publicly available datasets for plant disease detection.	[94]
6	Plant lowering	The robot comprised a sensing module, a robotic arm, a control box, and an operation module.	Achieved an average success rate of 60 % single roller hooks and 55 % for dual roller hooks.	[93]
7	Disease detection	Equipped with, MPU6050, SRF04 ultrasonic sensor, and Intel RealSense depth camera for SLAM tasks and tomato leaf imaging.	The ResNet-152 model achieved improved accuracy, precision, recall, and F1-score, compared to the PlantVillage dataset.	[95]
8	Greenhouse monitoring	Integrated RGB + NIR camera, GPS, and Wi-Fi	Mobile robotic platform enabled soil and canopy mapping inside greenhouses through visual and spectral data collection.	[96]

7. Overall summary

Recent research has extensively explored the use of cutting-edge technologies in greenhouses to enhance environmental control, disease detection, and crop monitoring. The review is divided into three sections. The first section (summarized in Table 1) focuses on the application of IoT in greenhouse systems. Most studies utilize sensors to collect environmental data—such as temperature, humidity, and soil moisture—to optimize plant growth conditions. The findings highlight the growing trend toward developing IoT-based automated or semi-

automated systems.

8. Conclusion

With increasing urbanization and the growing demand for efficient food production, innovative technologies have become essential in modern agriculture. Greenhouses play a key role in enhancing agricultural productivity and ensuring food security. This review highlights the significant contributions of IoT, ML/DL, and robotics to smart greenhouse systems. IoT enables real-time environmental monitoring and precise data collection, while ML/DL facilitates advanced analysis for disease detection, yield prediction, and climate optimization. Robotics further enhances automation by reducing human labor and improving operational accuracy in activities such as planting, harvesting, and spraying. The integration of these technologies has the potential to create intelligent, data-driven, and sustainable greenhouse management systems. However, challenges remain, such as high implementation costs, complex data integration, energy consumption, and the absence of established protocols. Future research should focus on developing robust AI models capable of operating in dynamic and unstructured greenhouse environments, along with affordable and lightweight substitutes. As technology continues to advance, the integration of IoT, AI, and robotics will play a significant role in shaping the next generation of smart greenhouses. This review consolidates recent studies on modern greenhouse systems, aiming to bridge the knowledge gap between intelligent technologies and plant biology to advance research on greenhouse monitoring and control. It provides technical insights into state-of-the-art intelligent systems for agricultural practitioners and researchers, encouraging continued exploration to fully harness the potential of AI and computer vision for next-generation smart greenhouse agriculture worldwide. The field holds countless opportunities and limitless possibilities for innovation and sustainable growth.

Funding

The Sultanate of Oman's Ministry of Energy and Minerals funded this research through committee members, including BP Oman, CC Energy Development Oman, OXY Oman, Daleel Petroleum, OQ, and Petroleum Development Oman. SQU also provided logistics support and in-kind contributions under the grant code CR/AGR/SWAE/23/03.

Ethics statement

Not applicable: This manuscript does not include human or animal research.

CRediT authorship contribution statement

Rawan Al-Najadi: Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Yaseen Al-Mulla:** Writing – review & editing, Writing – original draft, Conceptualization. **Khaled Goher:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

References

- [1] R. Rayhana, G. Xiao, Z. Liu, Internet of Things empowered smart greenhouse farming, *IEEE J. Radio Freq. Identif.* 4 (3) (2020) 195–211.
- [2] R.R. Shamshiri, F. Kalantari, K.C. Ting, K.R. Thorp, I.A. Hameed, C. Weltzien, Z. M. Shad, Advances in greenhouse automation and controlled environment agriculture: a transition to plant factories and urban agriculture, *Int. J. Agric. Biol. Eng.* 11 (1) (2018) 1–22.
- [3] K. Benke, B. Tomkins, Future food-production systems: vertical farming and controlled-environment agriculture, *Sustain.: Sci. Pract. Policy* 13 (1) (2017) 13–26.
- [4] Food and Agriculture Organization of the United Nations, *The Future of Food and Agriculture: Trends and Challenges*, FAO, 2017.
- [5] S. Abbate, P. Centobelli, R. Cerchione, The digital and sustainable transition of the agri-food sector, *Technol. Forecast. Soc. Change* 187 (2023) 122222.
- [6] B. Moura, H. Monteiro, T.M. Mata, M. Iten, A.A. Martins, Environmental life cycle assessment of early-stage development of ergosterol extraction from mushroom bio-residues, *J. Clean. Prod.* 355 (2022) 131623.
- [7] S. Testa, K.R. Nielsen, S. Vallentin, F. Ciccullo, Sustainability-oriented innovation in the agri-food system: current issues and the road ahead, *Technol. Forecast. Soc. Change* 179 (2022) 121653.
- [8] M. Pavleska, W.A. Kerr, Linking investment decisions and future food security to the regulation of genetically-based technologies, *Technol. Forecast. Soc. Change* 153 (2020) 119926.
- [9] A. Sagheer, M. Mohammed, K. Riad, M. Alhajhoj, A cloud-based IoT platform for precision control of soilless greenhouse cultivation, *Sensors* 21 (1) (2020) 223.
- [10] D.T. Santosh, C. Shukla, Innovations in greenhouse and controlled environment agriculture, *Innov. Hortic. Sustain. Growth* (2024).
- [11] M. Asaduzzaman, G. Niu, T. Asao, Nutrients recycling in hydroponics: opportunities and challenges toward sustainable crop production under controlled environment agriculture, *Front. Plant Sci.* 13 (2022) 845472.
- [12] T.D. Santosh, S. Maitra, Effect of drip irrigation and plastic mulch on yield and quality of ginger (*Zingiber officinale*), *Res. Crops* 23 (1) (2022) 211–219.
- [13] A.R. Chaurasia, *Population and Sustainable Development in India*, Springer, Singapore, 2020, pp. 1–10.
- [14] I. Iáñez-Rodríguez, M.Á. Martín-Lara, G. Blázquez, A. Pérez, M. Calero, Effect of torrefaction conditions on greenhouse crop residue: optimization of conditions to upgrade solid characteristics, *Bioresour. Technol.* 244 (2017) 741–749.
- [15] N. Singh, A.K. Sharma, I. Sarkar, S. Prabhu, K. Chadaga, IoT-based greenhouse technologies for enhanced crop production: a comprehensive study of monitoring, control, and communication techniques, *Syst. Sci. Control Eng.* 12 (1) (2024) 2306825.
- [16] S. Argento, G. Garcia, S. Treccarichi, Sustainable and low-input techniques in Mediterranean greenhouse vegetable production, *Horticulturae* 10 (9) (2024) 997.
- [17] A. Ouammi, Y. Achour, H. Dagdougui, D. Zejli, Optimal operation scheduling for a smart greenhouse integrated microgrid, *Energy Sustain. Dev.* 58 (2020) 129–137.
- [18] N. Choab, A. Allouhi, A. El Maakoul, T. Kousksou, S. Saadeddine, A. Jamil, Review on greenhouse microclimate and application: design parameters, thermal modeling and simulation, climate controlling technologies, *Sol. Energy* 191 (2019) 109–137.
- [19] R.K. Sahdev, M. Kumar, A.K. Dhingra, A comprehensive review of greenhouse shapes and its applications, *Front. Energy* 13 (2019) 427–438.
- [20] E. Cuce, D. Harjunowibowo, P.M. Cuce, Renewable and sustainable energy-saving strategies for greenhouse systems: a comprehensive review, *Renew. Sustain. Energy Rev.* 64 (2016) 34–59.
- [21] S. Gorjian, F. Calise, K. Kant, M.S. Ahamed, B. Copertaro, G. Najafi, R.R. Shamshiri, A review on opportunities for implementation of solar energy technologies in agricultural greenhouses, *J. Clean. Prod.* 285 (2021) 124807.
- [22] Y. Achour, A. Ouammi, D. Zejli, Technological progress in modern sustainable greenhouse cultivation is the path towards precision agriculture, *Renew. Sustain. Energy Rev.* 147 (2021) 111251.
- [23] R. Shamshiri, F. Kalantari, K.C. Ting, K.R. Thorp, I.A. Hameed, C. Weltzien, Z. M. Shad, Advances in greenhouse automation and controlled environment agriculture: a transition to plant factories and urban agriculture, *Int. J. Agric. Biol. Eng.* 11 (1) (2018) 1–22.
- [24] Y. Li, Y. Ding, D. Li, Z. Miao, Automatic carbon dioxide enrichment strategies in the greenhouse: a review, *Biosyst. Eng.* 171 (2018) 101–119.
- [25] M. Farvardin, M. Taki, S. Gorjian, E. Shabani, J.C. Sosa-Savedra, Assessing the physical and environmental aspects of greenhouse cultivation: a comprehensive review of conventional and hydroponic methods, *Sustainability* 16 (3) (2024) 1273.
- [26] N. Engler, M. Krarti, Review of energy efficiency in controlled environment agriculture, *Renew. Sustain. Energy Rev.* 141 (2021) 110786.
- [27] M. Zhang, T. Yan, W. Wang, X. Jia, J. Wang, J.J. Klemes, Energy-saving design and control strategy towards modern sustainable greenhouse: a review, *Renew. Sustain. Energy Rev.* 164 (2022) 112602.
- [28] R. Hesampour, M. Taki, R. Fathi, M. Hassani, A. Halog, Energy-economic-environmental cycle evaluation comparing two polyethylene and polycarbonate plastic greenhouses in cucumber production (from production to packaging and distribution), *Sci. Total Environ.* 828 (2022) 154232.
- [29] A. Badji, A. Benseddik, H. Bensaha, A. Boukhelifa, I. Hasrane, Design, technology, and management of greenhouse: a review, *J. Clean. Prod.* 373 (2022) 133753.
- [30] D. Garcia-Lesta, D. Cabello, E. Ferro, P. Lopez, V.M. Brea, Wireless sensor network with perpetual notes for terrestrial snail activity monitoring, *IEEE Sens., J* 17 (15) (2017) 5008–5015.

- [31] Y. Achour, A. Ouammi, D. Zejli, Technological progress in modern sustainable greenhouse cultivation is the path towards precision agriculture, *Renew. Sustain. Energy Rev.* 147 (2021) 111251.
- [32] G. Bagagiolo, G. Matrangola, E. Cavallo, N. Pampuro, Greenhouse robots: ultimate solutions to improve automation in protected cropping systems—A review, *Sustainability* 14 (11) (2022) 6436.
- [33] M.H. Saleem, J. Potgieter, K.M. Arif, Automation in agriculture by machine and deep learning techniques: a review of recent developments, *Precis. Agric.* 22 (6) (2021) 2053–2091.
- [34] C. Bersani, C. Ruggiero, R. Sacile, A. Soussi, E. Zero, Internet of things approaches for monitoring and control of smart greenhouses in industry 4.0, *Energies* 15 (10) (2022) 3834.
- [35] A. Boccalatte, M. Fossa, R. Sacile, Modeling, design, and construction of a zero-energy PV greenhouse for applications in Mediterranean climates, *Therm. Sci. Eng. Prog.* 25 (2021) 101046.
- [36] J. Contreras-Castillo, J.A. Guerrero-Ibañez, P.C. Santana-Mancilla, L. Anido-Rifon, SÁgriC-IoT: an IoT-based platform and deep learning for greenhouse monitoring, *Appl. Sci.* 13 (3) (2023) 1961.
- [37] M.S. Farooq, S. Riaz, A. Abid, K. Abid, M.A. Naeem, A survey on the role of IoT in agriculture for the implementation of smart farming, *IEEe Access.* 7 (2019) 156237–156271.
- [38] M.S. Farooq, R. Javid, S. Riaz, Z. Atal, IoT-based smart greenhouse framework and control strategies for sustainable agriculture, *IEEe Access.* 10 (2022) 99394–99420.
- [39] M.S. Farooq, S. Riaz, M.A. Helou, F.S. Khan, A. Abid, A. Alvi, Internet of things in greenhouse agriculture: a survey on enabling technologies, applications, and protocols, *IEEe Access.* 10 (2022) 53374–53397.
- [40] K.M. Hosny, W.M. El-Hady, F.M. Samy, Technologies, Protocols, and Applications of Internet of Things in Greenhouse Farming: A survey of Recent Advances, *Information Processing in Agriculture*, 2024.
- [41] C. Maraveas, D. Piromalis, K.G. Arvanitis, T. Bartzanas, D. Loukatos, Applications of IoT for optimized greenhouse environment and resource management, *Comput. Electron. Agric.* 198 (2022) 106993.
- [42] A. Sofwan, S. Sumardi, A.I. Ahmada, I. Ibrahim, K. Budiraharjo, K. Karno, Smart greethings: smart greenhouse based on internet of Things for environmental engineering, in: 2020 International Conference on Smart Technology and Applications (ICoSTA), IEEE, 2020, pp. 1–5.
- [43] S.J. Soheli, N. Jahan, M.B. Hossain, A. Adhikary, A.R. Khan, M. Wahiduzzaman, Smart greenhouse monitoring system using the internet of Things and artificial intelligence, *Wirel. Pers. Commun.* 124 (4) (2022) 3603–3634.
- [44] K. Tatas, A. Al-Zoubi, N. Christofides, C. Zannettis, M. Chrysostomou, S. Panteli, A. Antoniou, Reliable IoT-based monitoring and control of hydroponic systems, *Technologies* 10 (1) (2022) 26.
- [45] P. Kirci, E. Ozturk, Y. Celik, A novel approach for monitoring of smart greenhouse and flowerpot parameters and detection of plant growth with sensors, *Agriculture* 12 (10) (2022) 1705.
- [46] S. Ragaveena, A. Shirley Edward, U. Surendran, Smart controlled environment agriculture methods: a holistic review, *Rev. Environ. Sci. Bio/Technol.* 20 (4) (2021) 887–913.
- [47] S. Budiayanto, J.O. Manduapessy, L.M. Silalahi, A. Hamid, D. Hanafi, Wireless sensor network for power control and monitoring public street lighting with Earthquake early warning. 2024 FORTEI-International Conference on Electrical Engineering (FORTEI-ICEE), IEEE, 2024, pp. 235–240.
- [48] A.F. Subahi, K.E. Bouazza, An intelligent IoT-based system design for controlling and monitoring greenhouse temperature, *IEEe Access.* 8 (2020) 125488–125500.
- [49] J.S. Sujin, R. Murugan, M. Nagarjun, A.K. Praveen, IOT-based greenhouse monitoring and controlling system, in: *In Journal of Physics: Conference Series*, 1916, IOP Publishing, 2021 012062.
- [50] D.K. Widyawati, A. Ambarwari, A. Wahyudi, Design and prototype development of internet of things for greenhouse monitoring system, in: 2020 3rd International Seminar on Research of Information Technology and Intelligent Systems (ISRITI), IEEE, 2020, pp. 389–393.
- [51] Y. Yang, Design and application of intelligent agriculture service system with LoRa-based on wireless sensor network, in: 2020 International Conference on Computer Engineering and Application (ICCEA), IEEE, 2020, pp. 712–716.
- [52] H. Andrianto, A. Faizal, Development of smart greenhouse system for hydroponic agriculture. 2020 International Conference on Information Technology Systems and Innovation (ICITSI), IEEE, 2020, pp. 335–340.
- [53] Z. Sun, H. Yang, K. Liu, Z. Yin, Z. Li, W. Xu, Recent advances in LoRa: a comprehensive survey, *ACM. Trans. Sens. Netw.* 18 (4) (2022) 1–44.
- [54] S.O. Araújo, R.S. Peres, J. Barata, F. Lidon, J.C. Ramalho, Characterising the agriculture 4.0 landscape—Emerging trends, challenges and opportunities, *Agronomy* 11 (4) (2021) 667.
- [55] A. Harisha, C. Singh, S.S. US, K.T. Chethan, S. Hashim, H.G. Murdeshwar, AI controlled smart IoT-based greenhouse monitoring system, in: 2024 IEEE International Conference on Distributed Computing, VLSI, Electrical Circuits and Robotics (DISCOVER), IEEE, 2024, pp. 434–439.
- [56] J.S. Raj, J.V. Ananthi, Automation using IoT in greenhouse environment, *J. Inf. Technol.* 1 (01) (2019) 38–47.
- [57] H. Ibrahim, N. Mostafa, H. Halawa, M. Elsalamouny, R. Daoud, H. Amer, H. ElSayed, A layered IoT architecture for greenhouse monitoring and remote control, *SN Appl. Sci.* 1 (3) (2019) 223.
- [58] C. Kamiński, J.P. Soininen, M. Taumberger, R. Dantas, A. Toscano, T. Salmon Cinotti, A. Torre Neto, Smart water management platform: IoT-based precision irrigation for agriculture, *Sensors* 19 (2) (2019) 26.
- [59] P.P. Ray, Internet of things for smart agriculture: technologies, practices and future direction, *J Ambient Intell Smart Env.* 9 (4) (2017) 395–420.
- [60] M.A. Zamora-Izquierdo, J. Santa, J.A. Martínez, V. Martínez, A.F. Skarmeta, Smart farming IoT platform based on edge and cloud computing, *Biosyst. Eng.* 177 (2019) 4–17.
- [61] Adewumi, I.O. (2022). *Greenh. Monit.: Artif. Intell. (AI) Model Approach*.
- [62] M. Taki, S.A. Mehdizadeh, A. Rohani, M. Rahnama, M. Rahmati-Joneidabad, Applied machine learning in greenhouse simulation; new application and analysis, *Inf. Process. Agric.* 5 (2) (2018) 253–268.
- [63] J.U.M. Akbar, S.F. Kamarulzaman, A.J.M. Muzahid, M.A. Rahman, M. Uddin, A comprehensive review on deep learning assisted computer vision techniques for smart greenhouse agriculture, *IEEe Access.* 12 (2024) 4485–4522.
- [64] A. Rehman, J. Liu, L. Keqiu, A. Mateen, M.Q. Yasin, Machine learning prediction analysis using IoT for smart farming, *Int. J. 8* (9) (2020) 1–6.
- [65] M.O. Ojo, A. Zahid, Deep learning in controlled environment agriculture: a review of recent advancements, challenges and prospects, *Sensors* 22 (20) (2022) 7965.
- [66] C. Maraveas, Incorporating artificial intelligence technology in smart greenhouses: Current State of the art, *Appl. Sci.* 13 (1) (2022) 14.
- [67] M. Chen, X. Lang, X. Zhai, T. Li, Y. Shi, Intelligent recognition of greenhouse cucumber canopy vine top with deep learning model, *Comput. Electron. Agric.* 213 (2023) 108219.
- [68] M. Afonso, H. Fonteijn, F.S. Fiorentin, D. Lensink, M. Mooij, N. Faber, R. Wehrens, Tomato fruit detection and counting in greenhouses using deep learning, *Front. Plant Sci.* 11 (2020) 571299.
- [69] Y. Xu, Q. Chen, S. Kong, L. Xing, Q. Wang, X. Cong, Y. Zhou, Real-time object detection method of melon leaf diseases under complex background in greenhouse, *J. Real-Time Image Process.* 19 (5) (2022) 985–995.
- [70] S. Zhao, J. Liu, S. Wu, Multiple disease detection method for greenhouse-cultivated strawberry based on multiscale feature fusion faster R_CNN, *Comput. Electron. Agric.* 199 (2022) 107176.
- [71] P. Zhang, L. Yang, D. Li, EfficientNet-B4-Ranger: a novel method for greenhouse cucumber disease recognition under natural complex environment, *Comput. Electron. Agric.* 176 (2020) 105652.
- [72] V. Shukla, S. Rani, R.K. Mohapatra, A new approach for leaf disease detection using multilayered convolutional neural network, in: 2023 3rd International conference on Artificial Intelligence and Signal Processing (AISP), IEEE, 2023, pp. 1–5.
- [73] Y. Nan, H. Zhang, Y. Zeng, J. Zheng, Y. Ge, Intelligent detection of Multi-class pitaya fruits in target picking row based on WGB-YOLO network, *Comput. Electron. Agric.* 208 (2023) 107780.
- [74] N. Wang, T. Qian, J. Yang, L. Li, Y. Zhang, X. Zheng, J. Zhao, An enhanced YOLOv5 model for greenhouse cucumber fruit recognition based on color space features, *Agriculture* 12 (10) (2022) 1556.
- [75] R. Li, Z. Ji, S. Hu, X. Huang, J. Yang, W. Li, Tomato maturity recognition model based on improved yolov5 in greenhouse, *Agronomy* 13 (2) (2023) 603.
- [76] G. Liu, J.C. Nouaze, P.L. Touko Mbouembe, J.H. Kim, YOLO-tomato: a robust algorithm for tomato detection based on YOLOv3, *Sensors* 20 (7) (2020) 2145.
- [77] K. Liu, C. Zhang, X. Yang, M. Diao, H. Liu, M. Li, Development of an occurrence prediction model for cucumber downy mildew in solar greenhouses based on a long short-term memory neural network, *Agronomy* 12 (2) (2022) 442.
- [78] A. Badji, A. Benseddik, H. Bensaha, A. Boukhalifa, S. Bouhoun, C. Netti, D. Lalmi, Experimental assessment of a greenhouse with and without PCM thermal storage energy and prediction their thermal behavior using machine learning algorithms, *J. Energy Storage* 71 (2023) 108133.
- [79] S. Mao, Y. Li, Y. Ma, B. Zhang, J. Zhou, K. Wang, Automatic cucumber recognition algorithm for harvesting robots in the natural environment using deep learning and multi-feature fusion, *Comput. Electron. Agric.* 170 (2020) 105254.
- [80] G. Vijayakanthan, R. Vaishali, V. Abolghasemi, Detection of tea leaf diseases using deep transfer learning. 2024 Moratuwa Engineering Research Conference (MERCon), IEEE, 2024, pp. 13–18.
- [81] Proietti, M., Bianchi, F., Marini, A., Menculini, L., Termite, L.F., Garinei, A., ... & Marconi, M. (2021, April). Edge intelligence with Deep learning in Greenhouse Management. In *SMARTGREENS* (pp. 180–187).
- [82] M.E. Karar, F. Alsunaydi, S. Albusaymi, S. Alotaibi, A new mobile application of agricultural pests recognition using deep learning in cloud computing system, *Alex. Eng. J.* 60 (5) (2021) 4423–4432.
- [83] B.J. Cui, W.Q. Niu, Y.D. Du, Q. Zhang, Response of yield and nitrogen use efficiency to aerated irrigation and N application rate in greenhouse cucumber, *Sci. Hortic* 265 (2020) 109220.
- [84] J. Abdulridha, Y. Ampatzidis, P. Roberts, S.C. Kakarla, Detecting powdery mildew disease in squash at different stages using UAV-based hyperspectral imaging and artificial intelligence, *Biosyst. Eng.* 197 (2020) 135–148.
- [85] Y. Tang, M. Chen, C. Wang, L. Luo, J. Li, G. Lian, X. Zou, Recognition and localization methods for vision-based fruit picking robots: a review, *Front. Plant Sci.* 11 (2020) 510.
- [86] T. Kim, H. Park, J. Baek, M. Kim, D. Im, H. Park, D. Shin, Enhancement for Greenhouse sustainability using tomato disease image classification system based on intelligent complex controller, *Sustainability* 15 (23) (2023) 16220.
- [87] S. Blackmore, Towards robotic agriculture, in: *Autonomous Air and Ground Sensing Systems for Agricultural Optimization and Phenotyping*, 9866, SPIE, 2016, pp. 8–15.
- [88] Duckett, T., Pearson, S., Blackmore, S., Grieve, B., Chen, W.H., Cielniak, G., ... & Yang, G.Z. (2018). *Agricultural robotics: the future of robotic agriculture*. arXiv preprint arXiv:1806.06762.
- [89] G. Moreira, S.A. Magalhães, T. Pinho, F.N. dos Santos, M. Cunha, Benchmark of deep learning and a proposed HSV colour space models for the detection and classification of greenhouse tomato, *Agronomy* 12 (2) (2022) 356.

- [90] Y. Ge, S. Lin, Y. Zhang, Z. Li, H. Cheng, J. Dong, Z. Wu, Tracking and counting of tomato at different growth period using an improving YOLO-deepsort network for inspection robot, *Machines* 10 (6) (2022) 489.
- [91] B. Arad, J. Balendonck, R. Barth, O. Ben-Shahar, Y. Edan, T. Hellström, B. van Tuijl, Development of a sweet pepper harvesting robot, *J. Field Robot.* 37 (6) (2020) 1027–1039.
- [92] Y. Xiong, Y. Ge, L. Grimstad, P.J. From, An autonomous strawberry-harvesting robot: design, development, integration, and field evaluation, *J. Field Robot.* 37 (2) (2020) 202–224.
- [93] B. Zhang, S. Xu, Z. Xiong, H. Qin, X. Ai, T. Yuan, W. Li, Research on robot control technology of tomato plant lowering in greenhouses, *Agronomy* 14 (9) (2024) 1966.
- [94] N. Jadav, H. Chhajed, U. Patel, D. Jani, A. Barai, AI-enhanced quad-wheeled robot for targeted plant disease surveillance in greenhouses, in: *2023 Global Conference on Information Technologies and Communications (GCITC)*, IEEE, 2023, pp. 1–7.
- [95] T.T. Mac, T.D. Nguyen, H.K. Dang, D.T. Nguyen, X.T. Nguyen, Intelligent agricultural robotic detection system for greenhouse tomato leaf diseases using soft computing techniques and deep learning, *Sci. Rep.* 14 (1) (2024) 23887.
- [96] A. Milella, G. Reina, M. Nielsen, A multi-sensor robotic platform for ground mapping and estimation beyond the visible spectrum, *Precis. Agric.* 20 (2) (2019) 423–444.
- [97] F.K. Shaikh, S. Karim, S. Zeadally, J. Nebhen, Recent trends in internet-of-things-enabled sensor technologies for smart agriculture, *IEEE Internet Things J.* 9 (23) (2022) 23583–23598.
- [98] A. Rehman, T. Saba, M. Kashif, S.M. Fati, S.A. Bahaj, H. Chaudhry, A revisit of internet of things technologies for monitoring and control strategies in smart agriculture, *Agronomy* 12 (1) (2022) 127.
- [99] R. Khan, M.S. Farooq, A. Khelifi, U. Ahmad, F. Ahmad, S. Riaz, Internet of things (IoT) based saffron cultivation system in greenhouse, *Sci. Rep.* 14 (1) (2024) 22589.